

Contents

<i>Preface</i>	<i>vii</i>
1. Introduction	1
1.1 Macroscopic vs Microscopic Viewpoint	1
1.2 Thermodynamic System and Control Volume	2
1.3 Thermodynamic Properties, Processes and Cycles	3
1.4 Homogeneous and Heterogeneous Systems	4
1.5 Thermodynamic Equilibrium	4
1.6 Quasi-Static Process	5
1.7 Pure Substance	7
1.8 Concept of Continuum	7
1.9 Thermostatistics	8
1.10 Units and Dimensions	8
1.11 History of Thermodynamics	13
<i>Solved Examples</i>	20
<i>Review Questions</i>	21
<i>Problems</i>	22
2. Temperature	24
2.1 Zeroth Law of Thermodynamics	24
2.2 Measurement of Temperature—the Reference Points	24
2.3 Comparison of Thermometers	26
2.4 Ideal Gas	27
2.5 Gas Thermometers	28
2.6 Ideal Gas Temperature	29
2.7 Celsius Temperature Scale	30
2.8 Electrical Resistance Thermometer	30
2.9 Thermocouple	31
2.10 International Practical Temperature Scale	32
<i>Solved Examples</i>	33
<i>Review Questions</i>	35
<i>Problems</i>	35
3. Work and Heat Transfer	37
3.1 Work Transfer	37
3.2 $p dV$ -Work or Displacement Work	39
3.3 Indicator Diagram	42

3.4	Other Types of Work Transfer	44
3.5	Free Expansion with Zero Work Transfer	49
3.6	Net Work Done by a System	50
3.7	Heat Transfer	50
3.8	Heat Transfer—A Path Function	51
3.9	Specific Heat and Latent Heat	52
3.10	Point to Remember Regarding Heat Transfer and Work Transfer	53
	<i>Solved Examples</i>	54
	<i>Review Questions</i>	58
	<i>Problems</i>	59
4.	First Law of Thermodynamics	63
4.1	First Law for a Closed System Undergoing a Cycle	63
4.2	First Law for a Closed System Undergoing a Change of State	65
4.3	Energy—A Property of the System	66
4.4	Different Forms of Stored Energy	67
4.5	Specific Heat at Constant Volume	69
4.6	Enthalpy	70
4.7	Specific Heat at Constant Pressure	70
4.8	Energy of an Isolated System	71
4.9	Perpetual Motion Machine of the First Kind—PMM1	71
4.10	Limitations of the First Law	72
	<i>Solved Examples</i>	72
	<i>Review Questions</i>	77
	<i>Problems</i>	77
5.	First Law Applied to Flow Processes	81
5.1	Control Volume	81
5.2	Steady Flow Process	82
5.3	Mass Balance and Energy Balance in a Simple Steady Flow Process	83
5.4	Some Examples of Steady Flow Processes	86
5.5	Comparison of S.F.E.E. with Euler and Bernoulli Equations	90
5.6	Variable Flow Processes	91
5.7	Example of a Variable Flow Problem	93
5.8	Discharging and Charging a Tank	95
	<i>Solved Examples</i>	97
	<i>Review Questions</i>	106
	<i>Problems</i>	107
6.	Second Law of Thermodynamics	111
6.1	Qualitative Difference between Heat and Work	111
6.2	Cyclic Heat Engine	112
6.3	Energy Reservoirs	114
6.4	Kelvin-Planck Statement of Second Law	115

6.5	Clausius' Statement of the Second Law	116
6.6	Refrigerator and Heat Pump	116
6.7	Equivalence of Kelvin-Planck and Clausius Statements	119
6.8	Reversibility and Irreversibility	120
6.9	Causes of Irreversibility	121
6.10	Conditions for Reversibility	126
6.11	Carnot Cycle	126
6.12	Reversed Heat Engine	128
6.13	Carnot's Theorem	129
6.14	Corollary of Carnot's Theorem	131
6.15	Absolute Thermodynamic Temperature Scale	131
6.16	Efficiency of the Reversible Heat Engine	135
6.17	Equality of Ideal Gas Temperature and Kelvin Temperature	136
6.18	Types of Irreversibility	137
	<i>Solved Examples</i>	138
	<i>Review Questions</i>	146
	<i>Problems</i>	147

7. Entropy **152**

7.1	Introduction	152
7.2	Two Reversible Adiabatic Paths Cannot Intersect Each Other	152
7.3	Clausius' Theorem	153
7.4	The Property of Entropy	155
7.5	Principle of Caratheodory	158
7.6	The Inequality of Clausius	159
7.7	Entropy Change in an Irreversible Process	161
7.8	Entropy Principle	163
7.9	Applications of Entropy Principle	164
7.10	Entropy Transfer Mechanisms	171
7.11	Entropy Generation in a Closed System	173
7.12	Entropy Generation in an Open System	177
7.13	First and Second Laws Combined	178
7.14	Reversible Adiabatic Work in a Steady Flow System	179
7.15	Entropy and Direction: The Second Law—A Directional Law of Nature	182
7.16	Entropy and Disorder	182
7.17	Absolute Entropy	183
7.18	Entropy and Information Theory	183
7.19	Postulatory Thermodynamics	190
	<i>Solved Examples</i>	191
	<i>Review Questions</i>	204
	<i>Problems</i>	205

8. Available Energy, Exergy and Irreversibility **214**

8.1	Available Energy	214
8.2	Available Energy Referred to a Cycle	215

8.3	Quality of Energy	219
8.4	Maximum Work in a Reversible Process	222
8.5	Reversible Work by an Open System	
	Exchanging Heat only with the Surroundings	224
8.6	Useful Work	227
8.7	Dead State	229
8.8	Availability	230
8.9	Availability in Chemical Reactions	232
8.10	Irreversibility and Gouy-Stodola Theorem	234
8.11	Availability or Exergy Balance	237
8.12	Second Law Efficiency	240
8.13	Comments on Exergy	247
	<i>Solved Examples</i>	249
	<i>Review Questions</i>	271
	<i>Problems</i>	273
9.	Properties of Pure Substances	279
9.1	p - v Diagram for a Pure Substance	279
9.2	p - T Diagram for a Pure Substance	284
9.3	p - v - T Surface	285
9.4	T - s Diagram for a Pure Substance	285
9.5	h - s Diagram or Mollier Diagram for a Pure Substance	288
9.6	Quality or Dryness Fraction	291
9.7	Steam Tables	293
9.8	Charts of Thermodynamic Properties	295
9.9	Measurement of Steam Quality	295
	<i>Solved Examples</i>	302
	<i>Review Questions</i>	322
	<i>Problems</i>	322
10.	Properties of Gases and Gas Mixtures	328
10.1	Avogadro's Law	328
10.2	Equation of State of a Gas	328
10.3	Ideal Gas	331
10.4	Gas Compression	343
10.5	Equations of State	349
10.6	Virial Expansions	350
10.7	Law of Corresponding States	351
10.8	Other Equations of State	359
10.9	Properties of Mixtures of Gases—Dalton's Law of Partial Pressures	359
10.10	Internal Energy, Enthalpy and Specific Heats of Gas Mixtures	362
10.11	Entropy of Gas Mixtures	363
10.12	Gibbs Function of a Mixture of Inert Ideal Gases	365

<i>Solved Examples</i>	366
<i>Review Questions</i>	383
<i>Problems</i>	385

11. Thermodynamic Relations, Equilibrium and Third Law **396**

11.1	Some Mathematical Theorems	396
11.2	Maxwell's Equations	398
11.3	TdS Equations	399
11.4	Difference in Heat Capacities	399
11.5	Ratio of Heat Capacities	401
11.6	Energy Equation	402
11.7	Joule-Kelvin Effect	405
11.8	Clausius-Clapeyron Equation	409
11.9	Evaluation of Thermodynamic Properties from an Equation of State	412
11.10	General Thermodynamic Considerations on an Equation of State	415
11.11	Mixtures of Variable Composition	417
11.12	Conditions of Equilibrium of a Heterogeneous System	420
11.13	Gibbs Phase Rule	422
11.14	Types of Equilibrium	423
11.15	Local Equilibrium Conditions	426
11.16	Conditions of Stability	427
11.17	Third Law of Thermodynamics	428
	<i>Solved Examples</i>	434
	<i>Review Questions</i>	448
	<i>Problems</i>	449

12. Vapour Power Cycles **457**

12.1	Simple Steam Power Cycle	457
12.2	Rankine Cycle	459
12.3	Actual Vapour Cycle Processes	462
12.4	Comparison of Rankine and Carnot Cycles	464
12.5	Mean Temperature of Heat Addition	465
12.6	Reheat Cycle	468
12.7	Ideal Regenerative Cycle	470
12.8	Regenerative Cycle	472
12.9	Reheat-Regenerative Cycle	476
12.10	Feedwater Heaters	477
12.11	Exergy Analysis of Vapour Power Cycles	479
12.12	Characteristics of an Ideal Working Fluid in Vapour Power Cycles	481
12.13	Binary Vapour Cycles	483
12.14	Thermodynamics of Coupled Cycles	486
12.15	Process Heat and By-Product Power	487

Work and Heat Transfer

A closed system and its surroundings can interact in two ways: (a) by work transfer, and (b) by heat transfer. These may be called *energy interactions* and these bring about changes in the properties of the system. Thermodynamics mainly studies these energy interactions and the associated property changes of the system.

3.1 Work Transfer

Work is one of the basic modes of energy transfer. In mechanics the action of a force on a moving body is identified as work. A force is a means of transmitting an effect from one body to another. But a force itself never produces a physical effect except when coupled with motion and hence it is not a form of energy. An effect such as the raising of a weight through a certain distance can be performed by using a small force through a large distance or a large force through a small distance. The product of force and distance is the same to accomplish the same effect. In mechanics work is defined as:

The work is done by a force as it acts upon a body moving in the direction of the force.

The action of a force through a distance (or of a torque through an angle) is called *mechanical work* since other forms of work can be identified, as discussed later. The product of the force and the distance moved parallel to the force is the magnitude of mechanical work.

In thermodynamics, work transfer is considered as occurring between the system and the surroundings. *Work is said to be done by a system if the sole effect on things external to the system can be reduced to the raising of a weight.* The weight may not actually be raised, but the net effect external to the system would be the raising of a weight. Let us consider the battery and the motor in Fig. 3.1 as a system. The motor is driving a fan. The system is doing work upon

the surroundings. When the fan is replaced by a pulley and a weight, as shown in Fig. 3.2, the weight may be raised with the pulley driven by the motor. The sole effect on things external to the system is then the raising of a weight.

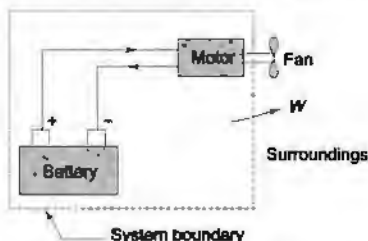


Fig. 3.1 Battery-motor system driving a fan

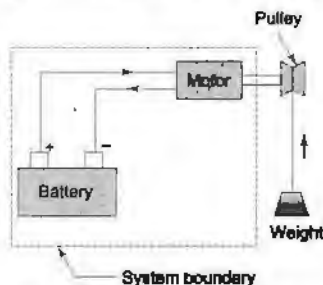


Fig. 3.2 Work transfer from a system

When work is done by a system, it is arbitrarily taken to be positive, and when work is done on a system, it is taken to be negative (Fig. 3.3.). The symbol W is used for work transfer.

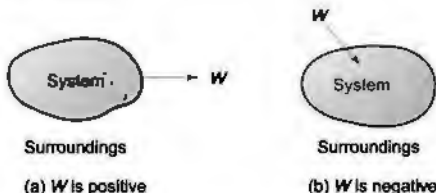


Fig. 3.3 Work interaction between a system and the surroundings

The unit of work is N.m or Joule [$1 \text{ Nm} = 1 \text{ Joule}$]. The rate at which work is done by, or upon, the system is known as *power*. The unit of power is J/s or watt.

Work is one of the forms in which a system and its surroundings can interact with each other. There are various types of work transfer which can get involved between them.

3.2 $p dV$ -Work or Displacement Work

Let the gas in the cylinder (Fig. 3.4) be a system having initially the pressure p_1 and volume V_1 . The system is in thermodynamic equilibrium, the state of which is described by the coordinates p_1 , V_1 . The piston is the only boundary which moves due to gas pressure. Let the piston move out to a new final position 2, which is also a thermodynamic equilibrium state specified by pressure p_2 and volume V_2 . At any intermediate point in the travel of the piston, let the pressure be p and the volume V . This must also be an equilibrium state, since macroscopic properties p and V are significant only for equilibrium states. When the piston moves an infinitesimal distance dl , and if ' a ' be the area of the piston, the force F acting on the piston $F = p \cdot a$, and the infinitesimal amount of work done by the gas on the piston

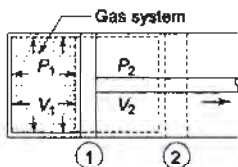


Fig. 3.4 $p dV$ work

$$\delta W = F \cdot dl = padl = pdV \quad (3.1)$$

where $dV = a dl$ = infinitesimal displacement volume. The differential sign in δW with the line drawn at the top of it will be explained later.

When the piston moves out from position 1 to position 2 with the volume changing from V_1 to V_2 , the amount of work W done by the system will be

$$W_{1-2} = \int_{V_1}^{V_2} p dV$$

The magnitude of the work done is given by the area under the path 1-2, as shown in Fig. 3.5. Since p is at all times a thermodynamic coordinate, all the states passed through by the system as the volume changes from V_1 to V_2 must be equilibrium states, and the path 1-2 must be **quasi-static**. The piston moves infinitely slowly so that

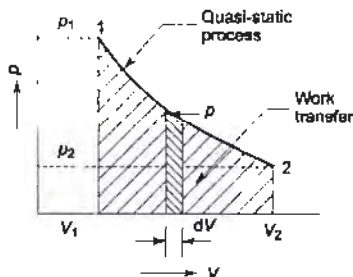


Fig. 3.5 Quasi-static $p dV$ work

every state passed through is an equilibrium state. The integration $\int p dV$ can be performed only on a quasi-static path.

3.2.1 Path Function and Point Function

With reference to Fig. 3.6, it is possible to take a system from state 1 to state 2 along many quasi-static paths, such as *A*, *B* or *C*. Since the area under each curve represents the work for each process, the amount of work involved in each case is not a function of the end states of the process, and it depends on the path the system follows in going from state 1 to state 2. For this reason, work is called a *path function*, and δW is an *inexact or imperfect differential*.

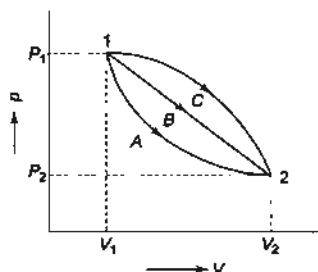


Fig. 3.6 Work—a path function

Thermodynamic properties are *point functions*, since for a given state, there is a definite value for each property. The change in a thermodynamic property of a system in a change of state is independent of the path the system follows during the change of state, and depends only on the initial and final states of the system. The differentials of point functions are *exact or perfect differentials*, and the integration is simply

$$\int_{V_1}^{V_2} dV = V_2 - V_1$$

The change in volume thus depends only on the end states of the system irrespective of the path the system follows.

On the other hand, work done in a quasi-static process between two given states depends on the path followed.

$$\int_1^2 \delta W \neq W_2 - W_1$$

Rather,

$$\int_1^2 \delta W = W_{1-2} \text{ or } {}_1W_2$$

To distinguish an inexact differential δW from an exact differential dV or dp the differential sign is being cut by a line at its top.

From Eq. (3.1),

$$dV = \frac{1}{p} dW \quad (3.2)$$

Here, $1/p$ is called the *integrating factor*. Therefore, an inexact differential dW when multiplied by an integrating factor $1/p$ becomes an exact differential dV .

For a cyclic process, the initial and final states of the system are the same, and hence, the change in any property is zero, i.e.

$$\oint dV = 0, \oint dp = 0, \oint dT = 0 \quad (3.3)$$

where the symbol $\oint$ denotes the cyclic integral for the closed path. Therefore, the cyclic integral of a property is always zero.

3.2.2 $p dV$ -Work in Various Quasi-Static Processes

(a) Constant pressure process (Fig. 3.7) (isobaric or isopiestic process)

$$W_{1-2} = \int_{V_1}^{V_2} p dV = p(V_2 - V_1) \quad (3.4)$$

(b) Constant volume process (Fig. 3.8) (isochoric process)

$$W_{1-2} = \int p dV = 0 \quad (3.5)$$

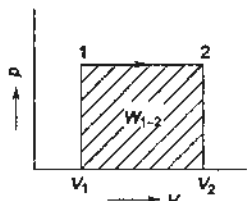


Fig. 3.7 Constant pressure process

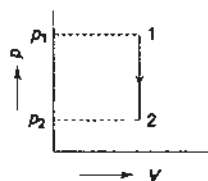


Fig. 3.8 Constant volume process

(c) Process in which $pV = C$ (Fig. 3.9)

$$\begin{aligned} \therefore W_{1-2} &= \int_{V_1}^{V_2} p dV, \quad pV = p_1 V_1 = C \\ p &= \frac{(p_1 V_1)}{V} \\ W_{1-2} &= p_1 V_1 \int_{V_1}^{V_2} \frac{dV}{V} = p_1 V_1 \ln \frac{V_2}{V_1} \\ &= p_1 V_1 \ln \frac{p_1}{p_2} \end{aligned} \quad (3.6)$$

(d) Process in which $pV^n = C$, where n is a constant (Fig. 3.10)

$$pV^n = p_1 V_1^n = p_2 V_2^n = C$$

$$\therefore p = \frac{(p_1 V_1^n)}{V^n}$$

$$\begin{aligned} \therefore W_{1-2} &= \int_{V_1}^{V_2} p dV \\ &= \int_{V_1}^{V_2} \frac{p_1 V_1^n}{V^n} \cdot dV \\ &= (p_1 V_1^n) \left[\frac{V^{-n+1}}{-n+1} \right]_{V_1}^{V_2} \\ &= \frac{p_1 V_1^n}{1-n} (V_2^{1-n} - V_1^{1-n}) \\ &= \frac{p_2 V_2^n \times V_2^{1-n} - p_1 V_1^n \times V_1^{1-n}}{1-n} \\ &= \frac{p_1 V_1 - p_2 V_2}{n-1} = \frac{p_1 V_1}{n-1} \left[1 - \left(\frac{p_2}{p_1} \right)^{n-1/n} \right] \quad (3.7) \end{aligned}$$

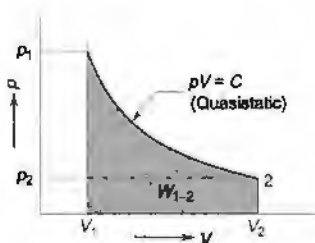


Fig. 3.9 Process in which $pV = \text{constant}$

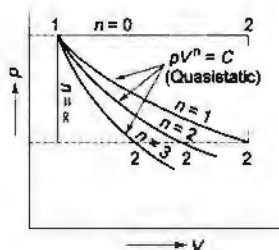


Fig. 3.10 Process in which $pV^n = \text{constant}$

3.3 Indicator Diagram

An indicator diagram is a trace made by a recording pressure gauge, called the indicator, attached to the cylinder of a reciprocating engine. This represents the work done in one engine cycle. Figure 3.11 shows a typical engine indicator.

The same gas pressure acts on both the engine piston P and the indicator piston I . The indicator piston is loaded by a spring and it moves in direct proportion to the change in pressure. The motion of the indicator piston causes a pencil held at the end of the linkage L to move upon a strip of paper wrapped around drum D . The drum is rotated about its axis by cord C , which is

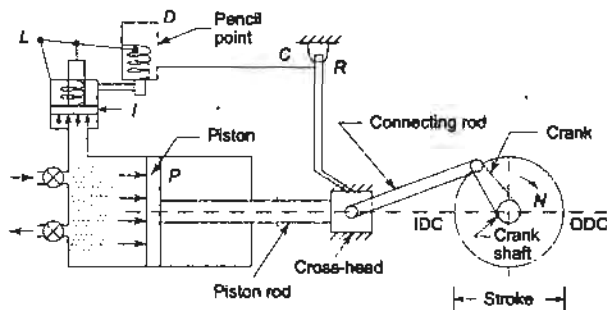


Fig. 3.11 Engine indicator

connected through a reducing motion R to the piston P of the engine. The surface of drum D moves horizontally under the pencil while the pencil moves vertically over the surface and a plot of pressure upon the piston vs. piston travel is obtained.

Before tracing the final indicator diagram, a pressure reference line is recorded by subjecting the indicator to the atmosphere and tracing a line at a constant pressure of one atmosphere.

The area of the indicator diagram represents the magnitude of the net work done by the system in one engine cycle. The area under the path 1-2 represents work done by the system and the area under the path 2-1 represents work done upon the system (Fig. 3.12). The area of the diagram, a_d , is measured by means of a planimeter, and the length of the diagram, l_d , is also measured. The mean effective pressure (m.e.p.) p_m is defined in the following way

$$p_m = \frac{a_d}{l_d} \times K$$

where K is the indicator spring constant ($\text{N/cm}^2 \times \text{cm travel}$). Work done in one engine cycle

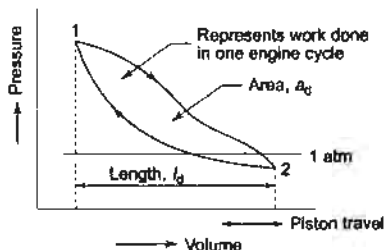


Fig. 3.12 Indicator diagram

$$= (p_m \cdot A) L$$

where A = cross-sectional area of the cylinder

$$= \frac{\pi}{4} D^2, \text{ where } D \text{ is the cylinder diameter}$$

and L = stroke of piston, or length of cylinder.

Let N be the revolutions per minute (r.p.m.) of the crankshaft. In a two stroke cycle, the engine cycle is completed in two strokes of the piston or in one revolution of the crankshaft. In a four-stroke cycle, the engine cycle is completed in four strokes of the piston or two revolutions of the crankshaft.

For a two-stroke engine, work done in one minute = $p_m ALN$, and for a four-stroke engine, work done in one minute = $p_m ALN/2$.

The power developed inside the cylinder of the engine is called *indicated power (IP)*,

$$\therefore \text{IP} = \frac{p_m AL \left(N \text{ or } \frac{N}{2} \right) n}{60} \text{ kW} \quad (3.8)$$

where p_m is in kPa and n is the number of cylinders in the engine.

The power available at the crankshaft is always less than this value (IP) due to friction, etc. and is called the *brake power (BP)* or *shaft power (SP)*. If ω is the angular velocity of the crankshaft in radian/sec, then

$$\text{BP} = T \omega \quad (3.9)$$

where T is the torque transmitted to the crankshaft in mN.

$$\therefore \text{BP} = \frac{2 \pi TN}{60} \quad (3.10)$$

where N is the number of revolutions per minute (rpm).

The mechanical efficiency of the engine, η_{mech} , is defined as

$$\eta_{\text{mech}} = \frac{\text{BP}}{\text{IP}} \quad (3.11)$$

An engine is said to be *double-acting*, if the working fluid is made to work on both sides of the piston. Such an engine theoretically develops twice the amount of work developed in a single-acting engine. Most reciprocating steam engines are double-acting, and so are many marine diesel engines. Internal combustion engines for road transport are always single-acting.

3.4 Other Types of Work Transfer

There are forms of work other than pdV or displacement work. The following are the additional types of work transfer which may get involved in system-surroundings interactions.

(a) **Electrical Work** When a current flows through a resistor (Fig. 3.13), taken as a system, there is work transfer into the system. This is because the current can drive a motor, the motor can drive a pulley and the pulley can raise a weight.

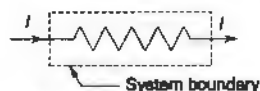


Fig. 3.13 Electrical work

The current flow, I , in amperes, is given by

$$I = \frac{dC}{d\tau}$$

where C is the charge in coulombs and τ is time in seconds. Thus dC is the charge crossing a boundary during time $d\tau$. If E is the voltage potential, the work is

$$\begin{aligned} dW &= E \cdot dC \\ &= EI d\tau \end{aligned}$$

$$\therefore W = \int_1^2 E I d\tau \quad (3.12)$$

The electrical power will be

$$\dot{W} = \lim_{d\tau \rightarrow 0} \frac{dW}{d\tau} = EI \quad (3.13)$$

This is the rate at which work is transferred.

(b) Shaft Work When a shaft, taken as the system (Fig. 3.14), is rotated by a motor, there is work transfer into the system. This is because the shaft can rotate a pulley which can raise a weight. If T is the torque applied to the shaft and $d\theta$ is the angular displacement of the shaft, the shaft work is

$$W = \int_1^2 T d\theta \quad (3.14)$$

and the shaft power is

$$\dot{W} = \int_1^2 T \frac{d\theta}{d\tau} = T \omega \quad (3.15)$$

where ω is the angular velocity and T is considered a constant in this case.

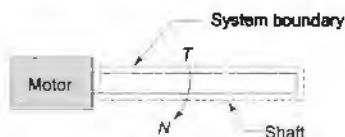


Fig. 3.14 Shaft work

(c) Paddle-Wheel Work or Stirring Work As the weight is lowered, and the paddle wheel turns (Fig. 3.15), there is work transfer into the fluid system

which gets stirred. Since the volume of the system remains constant, $\int p dV = 0$. If m is the mass of the weight lowered through a distance dz and T is the torque transmitted by the shaft in rotating through an angle $d\theta$, the differential work transfer to the fluid is given by

$$dW = mgdz = Td\theta$$

and the total work transfer is

$$W = \int_1^2 mgdz = \int_1^2 W' dz = \int_1^2 Td\theta \quad (3.15)$$

where W' is the weight lowered.

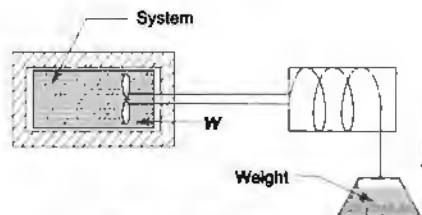


Fig. 3.15 Paddle-wheel work

(d) Flow Work The flow work, significant only in a flow process or an open system, represents the energy transferred across the system boundary as a result of the energy imparted to the fluid by a pump, blower or compressor to make the fluid flow across the control volume. Flow work is analogous to displacement work. Let p be the fluid pressure in the plane of the imaginary piston, which acts in a direction normal to it (Fig. 3.16). The work done on this imaginary piston by the external pressure as the piston moves forward is given by

$$\delta W_{\text{flow}} = p dV, \quad (3.16)$$

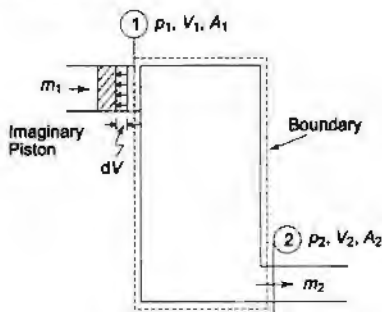


Fig. 3.16 Flow work

where dV is the volume of fluid element about to enter the system.

$$\therefore \delta W_{\text{flow}} = pv \, dm \quad (3.17)$$

where $dV = v \, dm$

Therefore, flow work at inlet (Fig. 3.16),

$$(\delta W_{\text{flow}})_{\text{in}} = p_1 v_1 \, dm_1 \quad (3.18)$$

Equation (3.18) can also be derived in a slightly different manner. If the normal pressure p_1 is exerted against the area A_1 , giving a total force $(p_1 A_1)$ against the piston, in time $d\tau$, this force moves a distance $V_1 d\tau$, where V_1 is the velocity of flow (piston). The work in time $d\tau$ is $p_1 A_1 V_1 d\tau$, or the work per unit time is $p_1 A_1 V_1$. Since the flow rate

$$w_1 = \frac{A_1 V_1}{v_1} = \frac{dm_1}{d\tau}$$

the work done in time $d\tau$ becomes

$$(\delta W_{\text{flow}})_{\text{in}} = p_1 v_1 \, dm_1$$

Similarly, flow work of the fluid element leaving the system is

$$(\delta W_{\text{flow}})_{\text{out}} = p_2 v_2 \, dm_2 \quad (3.19)$$

The flow work per unit mass is thus

$$W_{\text{flow}} = pv \quad (3.20)$$

It is the displacement work done at the moving system boundary.

(e) Work Done in Stretching a Wire Let us consider a wire as the system. If the length of the wire in which there is a tension $\mathcal{F}$ is changed from L to $L + dL$, the infinitesimal amount of work that is done is equal to

$$\delta W = -\mathcal{F} \, dL$$

The minus sign is used because a positive value of dL means an expansion of the wire, for which work must be done on the wire, i.e., negative work. For a finite change of length,

$$W = - \int_1^2 \mathcal{F} \, dL \quad (3.21)$$

If we limit the problem to within the elastic limit, where E is the modulus of elasticity, s is the stress, ϵ is the strain, and A is the cross-sectional area, then

$$\mathcal{F} = sA = E\epsilon A, \text{ since } \frac{s}{\epsilon} = E$$

$$d\epsilon = \frac{dL}{L}$$

$$\delta W = -\mathcal{F} \, dL = -E\epsilon AL \, d\epsilon$$

$$\therefore W = -A\epsilon L \int_1^2 \epsilon \, d\epsilon = -\frac{AEL}{2} (\epsilon_2^2 - \epsilon_1^2) \quad (3.22)$$

(f) Work Done in Changing the Area of a Surface Film A film on the surface of a liquid has a surface tension, which is a property of the liquid and the surroundings. The surface tension acts to make the surface area of the liquid a minimum. It has the unit of force per unit length. The work done on a homogeneous liquid film in changing its surface area by an infinitesimal amount dA is

$$dW = -\sigma dA$$

where σ is the surface tension (N/m).

$$\therefore W = - \int_1^2 \sigma dA \quad (3.23)$$

(g) Magnetization of a Paramagnetic Solid The work done per unit volume on a magnetic material through which the magnetic and magnetization fields are uniform is

$$dW = -HdI$$

$$\text{and} \quad W_{1-2} = - \int_{I_1}^{I_2} HdI \quad (3.24)$$

where H is the field strength, and I is the component of the magnetization field in the direction of the field. The minus sign provides that an increase in magnetization (positive dI) involves negative work.

The following equations summarize the different forms of work transfer:

Displacement work

$$\text{(compressible fluid)} \quad W = \int_1^2 p dV$$

$$\text{Electrical work} \quad W = \int_1^2 EdC = \int_1^2 EI \, d\tau$$

$$\text{Shaft work} \quad W = \int_1^2 T d\theta$$

$$\text{Surface film} \quad W = - \int_1^2 \sigma dA \quad (3.25)$$

$$\text{Stretched wire} \quad W = - \int_1^2 \mathcal{F} dL$$

Magnetised solid	$W = - \int_1^2 H \, dI$
------------------	--------------------------

It may be noted in the above expressions that the work is equal to the integral of the product of an intensive property and the change in its related extensive property. These expressions are valid only for infinitesimally slow quasi-static processes.

There are some other forms of work which can be identified in processes that are not quasi-static, for example, the work done by shearing forces in a process involving friction in a viscous fluid.

3.5 Free Expansion with Zero Work Transfer

Work transfer is identified only at the boundaries of a system. It is a boundary phenomenon, and a form of energy in transit crossing the boundary. Let us consider a gas separated from the vacuum by a partition (Fig. 3.17). Let the partition be removed. The gas rushes to fill the entire volume. The expansion of a gas against vacuum is called *free expansion*. If we neglect the work associated with the removal of partition, and consider the gas and vacuum together as our system (Fig. 3.17a), there is no work transfer involved here, since no work crosses the system boundary, and hence

$$\int_1^2 \delta W = 0, \text{ although } \int_1^2 p \, dV \neq 0$$

If only the gas is taken as the system (Fig. 3.17), when the partition is removed there is a change in the volume of the gas, and one is tempted to calculate the work from the expression $\int_1^2 p \, dV$. However, this is not a quasistatic process, although the initial and final end states are in equilibrium. Therefore, the work cannot be calculated from this relation. The two end states can be located on the p - V diagram and these are joined by a dotted line (Fig. 3.17c) to indicate that the process had occurred. However, if the vacuum space is divided into a large number of small volumes by partitions and the partitions are removed one by one slowly (Fig. 3.17d), then every state passed through by the system is an equilibrium state and the work done can then be estimated from the relation $\int_1^2 p \, dV$ (Fig. 3.17e). Yet, in free expansion of a gas, there is no resistance to the fluid at the system boundary as the volume of the gas increases to fill up the vacuum space. Work is done by a system to overcome some resistance. Since vacuum does not offer any resistance, there is no work transfer involved in free expansion.

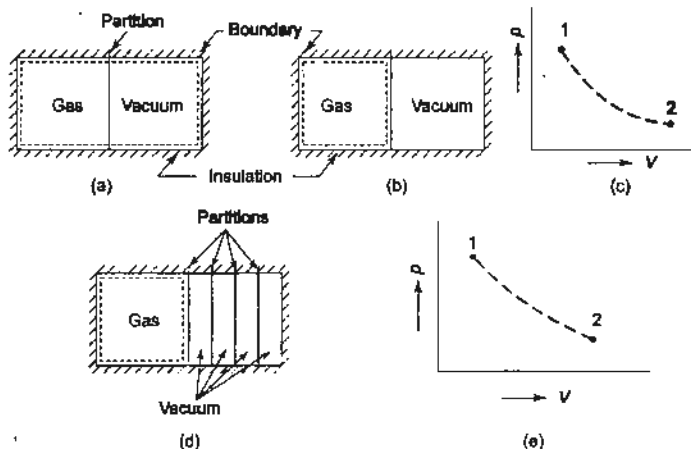


Fig. 3.17 Free expansion

3.6 Net Work Done by a System

Often different forms of a work transfer occur simultaneously during a process executed by a system. When all these work interactions have been evaluated, the total or net work done by the system would be equal to the algebraic sum of these as given below

$$W_{\text{total}} = W_{\text{displacement}} + W_{\text{shear}} + W_{\text{electrical}} + W_{\text{stirring}} + \dots$$

3.7 Heat Transfer

Heat is defined as the form of energy that is transferred across a boundary by virtue of a temperature difference. The temperature difference is the 'potential' or 'force' and heat transfer is the 'flux'.

The transfer of heat between two bodies in direct contact is called *conduction*. Heat may be transferred between two bodies separated by empty space or gases by the mechanism of *radiation* through electromagnetic waves. A third method of heat transfer is *convection* which refers to the transfer of heat between a wall and a fluid system in motion.

The direction of heat transfer is taken from the high temperature system to the low temperature system. *Heat flow into a system is taken to be positive, and heat flow out of a system is taken as negative* (Fig. 3.13). The symbol Q is used for heat transfer, i.e., the quantity of heat transferred within a certain time.

Heat is a form of energy in transit (like work transfer). It is a boundary phenomenon, since it occurs only at the boundary of a system. Energy transfer by virtue of temperature difference only is called heat transfer. All other energy interactions may be termed as work transfer.

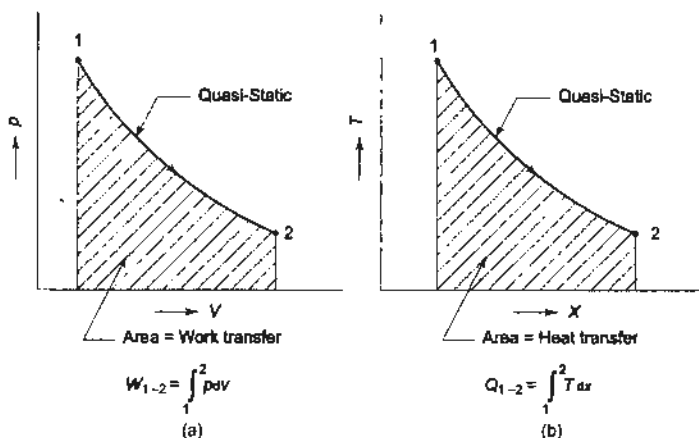


Fig. 3.19 Representation of work transfer and heat transfer in quasi-static processes on p - v and T - x coordinates

Just like displacement work, the heat transfer can also be written as the integral of the product of the intensive property T and the differential change of an extensive property, say X (Fig. 3.19b).

$$Q_{1-2} = \int_1^2 \delta Q = \int_1^2 T dX \quad (3.26)$$

It must also be valid for a quasi-static process only, and the heat transfer involved is represented by the area under the path 1-2 in T - X plot (Fig. 3.19b). Heat transfer is, therefore, a path function, i.e., the amount of heat transferred when a system changes from a state 1 to a state 2 depends on the path the system follows (Fig. 3.19b). Therefore, δQ is an inexact differential. Now,

$$\delta Q = T dX$$

where X is an extensive property and dX is an exact differential.

$$\therefore dX = \frac{1}{T} \delta Q \quad (3.27)$$

To make δQ integrable, i.e., an exact differential, it must be multiplied by an integrating factor which is, in this case, $1/T$. The extensive property X is yet to be defined. It has been introduced in Chapter 7 and it is called 'entropy'.

3.9 Specific Heat and Latent Heat

The *specific heat* of a substance is defined as the amount of heat required to raise a unit mass of the substance through a unit rise in temperature. The symbol c will be used for specific heat.

$$\therefore c = \frac{Q}{m \cdot \Delta t} \text{ J/kg K}$$

where Q is the amount of heat transfer (J), m , the mass of the substance (kg), and Δt , the rise in temperature (K).

Since heat is not a property, as explained later, so the specific heat is qualified with the process through which exchange of heat is made. For gases, if the process is at constant pressure, it is c_p , and if the process is at constant volume, it is c_v . For solids and liquids, however, the specific heat does not depend on the process. An elegant manner of defining specific heats, c_v and c_p , in terms of properties is given in Secs 4.5 and 4.6.

The product of mass and specific heat (mc) is called the *heat capacity* of the substance. The capital letter C , C_p or C_v , is used for heat capacity.

The latent heat is the amount of heat transfer required to cause a phase change in unit mass of a substance *at a constant pressure and temperature*. There are three phases in which matter can exist: solid, liquid, and vapour or gas. The *latent heat of fusion* (l_{fu}) is the amount of heat transferred to melt unit mass of solid into liquid, or to freeze unit mass of liquid to solid. The *latent heat of vaporization* (l_{vap}) is the quantity of heat required to vaporize unit mass of liquid into vapour, or condense unit mass of vapour into liquid. The *latent heat of sublimation* (l_{sub}) is the amount of heat transferred to convert unit mass of solid to vapour or vice versa. l_{fu} is not much affected by pressure, whereas l_{vap} is highly sensitive to pressure.

3.10 Points to Remember Regarding Heat Transfer and Work Transfer

- Heat transfer and work transfer are the *energy interactions*. A closed system and its surroundings can interact in two ways: by heat transfer and by work transfer. Thermodynamics studies how these interactions bring about property changes in a system.
- The same effect in a closed system can be brought about either by heat transfer or by work transfer. Whether heat transfer or work transfer has taken place depends on what constitutes the system.
- Both heat transfer and work transfer are boundary phenomena. Both are observed at the boundaries of the system, and both represent energy crossing the boundaries of the system.
- It is wrong to say 'total heat' or 'heat content' of a closed system, because heat or work is not a property of the system. Heat, like work, cannot be stored by the system. Both heat and work are the energy in transit.
- Heat transfer is the energy interaction due to temperature difference only. All other energy interactions may be termed as work transfer.
- Both heat and work are path functions and inexact differentials. The magnitude of heat transfer or work transfer depends upon the path the system follows during the change of state.

SOLVED EXAMPLES

Example 3.1 Gas from a bottle of compressed helium is used to inflate an inelastic flexible balloon, originally folded completely flat to a volume of 0.5 m^3 . If the barometer reads 760 mm Hg , what is the amount of work done upon the atmosphere by the balloon? Sketch the system before and after the process.

Solution The firm line P_1 (Fig. Ex. 3.1) shows the boundary of the system before the process, and the dotted line P_2 shows the boundary after the process. The displacement work

$$\begin{aligned} W_d &= \int_{\text{Balloon}} p dV + \int_{\text{Bottle}} p dV = p \Delta V + 0 \\ &= 101.325 \frac{\text{kN}}{\text{m}^2} \times 0.5 \text{ m}^3 \\ &= 50.66 \text{ kJ} \end{aligned}$$

This is positive because work is done by the system. Work done by the atmosphere is -50.66 kJ . Since the wall of the bottle is rigid, there is no $p dV$ -work involved in it.

It is assumed that the pressure in the balloon is atmospheric at all times, since the balloon fabric is light, inelastic and unstressed. If the balloon were elastic and stressed during the filling process, the work done by the gas would be greater than 50.66 kJ by an amount equal to the work done in stretching the balloon, although the displacement work done by the atmosphere is still -50.66 kJ . However, if the system includes both the gas and the balloon, the displacement work would be 50.66 kJ , as estimated above.

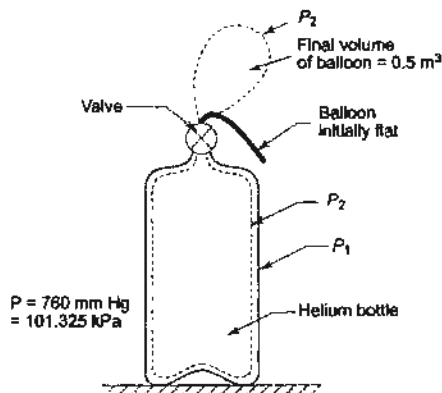


Fig. Ex. 3.1

Example 3.2 When the value of the evacuated bottle (Fig. Ex. 3.2) is opened, atmospheric air rushes into it. If the atmospheric pressure is 101.325 kPa, and 0.6 m³ of air (measured at atmospheric conditions) enters into the bottle, calculate the work done by air.

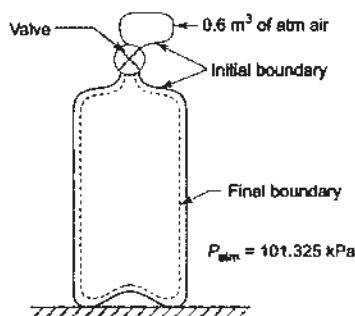


Fig. Ex. 3.2

Solution The displacement work done by air

$$\begin{aligned}
 W_d &= \int_{\text{Bottle}} p dV + \int_{\text{Free-air boundary}} p dV \\
 &= 0 + p \Delta V \\
 &= 101.325 \text{ kN/m}^2 \times 0.6 \text{ m}^3 \\
 &= 60.8 \text{ kJ}
 \end{aligned}$$

Since the free-air boundary is contracting, the work done by the system is negative (ΔV being negative), and the surroundings do positive work upon the system.

Example 3.3 A piston and cylinder machine containing a fluid system has a stirring device in the cylinder (Fig. Ex. 3.3). The piston is frictionless, and it is held down against the fluid due to the atmospheric pressure of 101.325 kPa. The stirring device is turned 10,000 revolutions with an average torque against the fluid of 1.275 mN. Meanwhile the piston of 0.6 m diameter moves out 0.8 m. Find the net work transfer for the system.

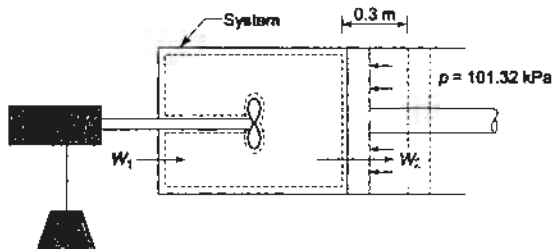


Fig. Ex. 3.3

Solution Work done by the stirring device upon the system (Fig. Ex. 3.3).

$$\begin{aligned} W_1 &= 2\pi TN \\ &= 2\pi \times 1.275 \times 10,000 \text{ Nm} \\ &= 80 \text{ kJ} \end{aligned}$$

This is negative work for the system.

Work done by the system upon the surroundings

$$\begin{aligned} W_2 &= (pA) \cdot L \\ &= 101.325 \frac{\text{kN}}{\text{m}^2} \times \frac{\pi}{4} (0.6)^2 \text{ m}^2 \times 0.80 \text{ m} \\ &= 22.9 \text{ kJ} \end{aligned}$$

This is positive work for the system. Hence, the net work transfer for the system

$$W = W_1 + W_2 = -80 + 22.9 = -57.1 \text{ kJ}$$

Example 3.4 The following data refer to a 12-cylinder, single-acting, two-stroke marine diesel engine:

Speed—150 rpm

Cylinder diameter—0.8 m

Stroke of piston—1.2 m

Area of indicator diagram— $5.5 \times 10^{-4} \text{ m}^2$

Length of diagram—0.06 m

Spring value—147 MPa per m

Find the net rate of work transfer from the gas to the pistons in kW.

Solution Mean effective pressure, p_m , is given by

$$\begin{aligned} p_m &= \frac{a_d}{l_d} \times \text{spring constant} \\ &= \frac{5.5 \times 10^{-4} \text{ m}^2}{0.06} \times 147 \frac{\text{MPa}}{\text{m}} \\ &= 1.35 \text{ MPa} \end{aligned}$$

One engine cycle is completed in two strokes of the piston or one revolution of the crank-shaft.

$\therefore$ Work done in one minute

$$\begin{aligned} &= p_m LAN \\ &= 1.35 \times \frac{\pi}{4} (0.8)^2 \times 1.2 \times 150 = 122 \text{ MJ} \end{aligned}$$

Since the engine is single-acting, and it has 12 cylinders, each contributing an equal power, the rate of work transfer from the gas to the piston is given by

$$\begin{aligned} W &= 122 \times 12 \text{ MJ/min} \\ &= 24.4 \text{ MJ/s} \\ &= 24.4 \text{ MW} = 24,400 \text{ kW} \end{aligned}$$

Ans.

Example 3.5 It is required to melt 5 tonnes/h of iron from a charge at 15°C to molten metal at 1650°C . The melting point is 1535°C , and the latent heat is 270 kJ/kg . The specific heat in solid state is 0.502 and in liquid state ($29.93/\text{atomic weight}$) kJ/kg K . If an electric furnace has 70% efficiency, find the kW rating needed. If the density in molten state is 6900 kg/m^3 and the bath volume is three times the hourly melting rate, find the dimensions of the cylindrical furnace if the length to diameter ratio is 2. The atomic weight of iron is 56.

Solution Heat required to melt 1 kg of iron at 15°C to molten metal at 1650°C
 $=$ Heat required to raise the temperature from 15°C to 1535°C
 $+ \text{ Latent heat } + \text{ Heat required to raise the temperature from } 1535^{\circ}\text{C to } 1650^{\circ}\text{C}$
 $= 0.502 (1535 - 15) + 270 + 29.93 (1650 - 1535)/56$
 $= 763 + 270 + 61.5$
 $= 1094.5 \text{ kJ/kg}$

Melting rate $= 5 \times 10^3 \text{ kg/h}$

So, the rate of heat supply required

$$= (5 \times 10^3 \times 1094.5) \text{ kJ/h}$$

Since the furnace has 70% efficiency, the rating of the furnace would be

$$= \frac{\text{Rate of heat supply per second}}{\text{Furnace efficiency}}$$

$$= \frac{5 \times 10^3 \times 1094.5}{0.7 \times 3600} = 217 \times 10^3 \text{ kW}$$

Ans.

$$\text{Volume needed} = \frac{3 \times 5 \times 10^3}{6900} \text{ m}^3 = 2.18 \text{ m}^3$$

If d is the diameter and l the length of the furnace

$$\frac{\pi}{4} d^2 l = 2.18 \text{ m}^3$$

$$\text{or} \quad \frac{\pi}{4} d^2 \times 2d = 2.18 \text{ m}^3$$

$$\therefore d = 1.15 \text{ m}$$

$$\text{and} \quad l = 2.30 \text{ m}$$

Ans.

Example 3.6 If it is desired to melt aluminium with solid state specific heat 0.9 kJ/kgK , latent heat 390 kJ/kg , atomic weight 27, density in molten state 2400 kg/m^3 and final temperature 700°C , find out how much metal can be melted per hour with the above kW rating. Other data are as in the above example. Also, find the mass of aluminium that the above furnace will hold. The melting point of aluminium is 660°C .

Solution Heat required per kg of aluminium

$$\begin{aligned}
 &= 0.9(660 - 15) + 390 + \frac{29.93}{27}(700 - 660) \\
 &= 580.5 + 390 + 44.3 \\
 &= 1014.8 \text{ kJ}
 \end{aligned}$$

$$\text{Heat to be supplied} = \frac{1014.8}{0.7} = 1449.7 \text{ kJ/kg}$$

With the given power, the rate at which aluminium can be melted

$$\begin{aligned}
 &= \frac{2.17 \times 10^3 \times 3600}{1449.7} \text{ kg/h} \\
 &= 5.39 \text{ tonnes/h}
 \end{aligned}$$

Ans.

Mass of aluminium that can be held in the above furnace

$$\begin{aligned}
 &= 2.18 \times 2400 \text{ kg} \\
 &= 5.23 \text{ tonnes}
 \end{aligned}$$

Ans.

REVIEW QUESTIONS

- 3.1 How can a closed system and its surroundings interact? What is the effect of such interactions on the system?
- 3.2 When is work said to be done by a system?
- 3.3 What are positive and negative work interactions?
- 3.4 What is displacement work?
- 3.5 Under what conditions is the work done equal to $\int_1^2 p \, dv$?
- 3.6 What do you understand by path function and point function? What are exact and inexact differentials?
- 3.7 Show that work is a path function, and not a property.
- 3.8 What is an indicator diagram?
- 3.9 What is mean effective pressure? How is it measured?
- 3.10 What are the indicated power and the brake power of an engine?
- 3.11 How does the current flowing through a resistor represent work transfer?
- 3.12 What do you understand by flow work? Is it different from displacement work?
- 3.13 Why does free expansion have zero work transfer?
- 3.14 What is heat transfer? What are its positive and negative directions?
- 3.15 What are adiabatic and diathermic walls?
- 3.16 What is an integrating factor?
- 3.17 Show that heat is a path function and not a property.
- 3.18 What is the difference between work transfer and heat transfer?
- 3.19 Does heat transfer inevitably cause a temperature rise?

PROBLEMS

- 3.1 (a) A pump forces $1 \text{ m}^3/\text{min}$ of water horizontally from an open well to a closed tank where the pressure is 0.9 MPa . Compute the work the pump must do upon the water in an hour just to force the water into the tank against the pressure. Sketch the system upon which the work is done before and after the process.
- Ans.* 13.31 kJ
- (b) If the work done as above upon the water had been used solely to raise the same amount of water vertically against gravity without change of pressure, how many meters would the water have been elevated?
- (c) If the work done in (a) upon the water had been used solely to accelerate the water from zero velocity without change of pressure or elevation, what velocity would the water have reached? If the work had been used to accelerate the water from an initial velocity of 10 m/s , what would the final velocity have been?
- 3.2 The piston of an oil engine, of area 0.0045 m^2 , moves downwards 75 mm , drawing in 0.00028 m^3 of fresh air from the atmosphere. The pressure in the cylinder is uniform during the process at 80 kPa , while the atmospheric pressure is 101.325 kPa , the difference being due to the flow resistance in the induction pipe and the inlet valve. Estimate the displacement work done by the air finally in the cylinder.
- Ans.* 27 J
- 3.3 An engine cylinder has a piston of area 0.12 m^2 and contains gas at a pressure of 1.5 MPa . The gas expands according to a process which is represented by a straight line on a pressure-volume diagram. The final pressure is 0.15 MPa . Calculate the work done by the gas on the piston if the stroke is 0.30 m .
- Ans.* 29.7 kJ
- 3.4 A mass of 1.5 kg of air is compressed in a quasi-static process from 0.1 MPa to 0.7 MPa for which $pv = \text{constant}$. The initial density of air is 1.16 kg/m^3 . Find the work done by the piston to compress the air.
- Ans.* 251.62 kJ
- 3.5 A mass of gas is compressed in a quasi-static process from 80 kPa , 0.1 m^3 to 0.4 MPa , 0.03 m^3 . Assuming that the pressure and volume are related by $pv^n = \text{constant}$, find the work done by the gas system.
- Ans.* -11.83 kJ
- 3.6 A single-cylinder, double-acting, reciprocating water pump has an indicator diagram which is a rectangle 0.075 m long and 0.05 m high. The indicator spring constant is 147 MPa per m . The pump runs at 50 rpm . The pump cylinder diameter is 0.15 m and the piston stroke is 0.20 m . Find the rate in kW at which the piston does work on the water.
- Ans.* 43.3 kW
- 3.7 A single-cylinder, single-acting, 4 stroke engine of 0.15 m bore develops an indicated power of 4 kW when running at 216 rpm . Calculate the area of the indicator diagram that would be obtained with an indicator having a spring constant of $25 \times 10^6 \text{ N/m}^3$. The length of the indicator diagram is 0.1 times the length of the stroke of the engine.

Ans. 505 mm^2

- 3.8 A six-cylinder, 4-stroke gasoline engine is run at a speed of 2520 RPM. The area of the indicator card of one cylinder is $2.45 \times 10^3 \text{ mm}^2$ and its length is 58.5 mm. The spring constant is $20 \times 10^6 \text{ N/m}^3$. The bore of the cylinders is 140 mm and the piston stroke is 150 mm. Determine the indicated power, assuming that each cylinder contributes an equal power.

Ans. 243.57 kW

- 3.9 A closed cylinder of 0.25 m diameter is fitted with a light frictionless piston. The piston is retained in position by a catch in the cylinder wall and the volume on one side of the piston contains air at a pressure of 750 kN/m^2 . The volume on the other side of the piston is evacuated. A helical spring is mounted coaxially with the cylinder in this evacuated space to give a force of 120 N on the piston in this position. The catch is released and the piston travels along the cylinder until it comes to rest after a stroke of 1.2 m. The piston is then held in its position of maximum travel by a ratchet mechanism. The spring force increases linearly with the piston displacement to a final value of 5 kN. Calculate the work done by the compressed air on the piston.

Ans. 3.07 kJ

- 3.10 A steam turbine drives a ship's propeller through an 8 : 1 reduction gear. The average resisting torque imposed by the water on the propeller is $750 \times 10^3 \text{ N}$ and the shaft power delivered by the turbine to the reduction gear is 15 MW. The turbine speed is 1450 rpm. Determine (a) the torque developed by the turbine, (b) the power delivered to the propeller shaft, and (c) the net rate of working of the reduction gear.

Ans. (a) $T = 98.84 \text{ kN m}$, (b) 14.235 MW, (c) 0.765 MW

- 3.11 A fluid, contained in a horizontal cylinder fitted with a frictionless leakproof piston, is continuously agitated by means of a stirrer passing through the cylinder cover. The cylinder diameter is 0.40 m. During the stirring process lasting 10 minutes, the piston slowly moves out a distance of 0.485 m against the atmosphere. The net work done by the fluid during the process is 2 kJ. The speed of the electric motor driving the stirrer is 840 rpm. Determine the torque in the shaft and the power output of the motor.

Ans. 0.08 mN, 6.92 W

- 3.12 At the beginning of the compression stroke of a two-cylinder internal combustion engine the air is at a pressure of 101.325 kPa. Compression reduces the volume to 1/5 of its original volume, and the law of compression is given by $pV^{1.2} = \text{constant}$. If the bore and stroke of each cylinder is 0.15 m and 0.25 m, respectively, determine the power absorbed in kW by compression strokes when the engine speed is such that each cylinder undergoes 500 compression strokes per minute.

Ans. 17.95 kW

- 3.13 Determine the total work done by a gas system following an expansion process as shown in Fig. P. 3.13.

Ans. 2952 MJ

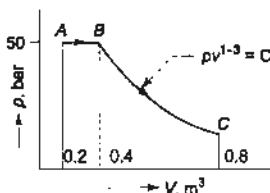


Fig. P. 3.13

- 3.14 A system of volume V contains a mass m of gas at pressure p and temperature T . The macroscopic properties of the system obey the following relationship:

$$\left(p + \frac{a}{V^2}\right)(V - b) = mRT$$

where a , b , and R are constants.

Obtain an expression for the displacement work done by the system during a constant-temperature expansion from volume V_1 to volume V_2 . Calculate the work done by a system which contains 10 kg of this gas expanding from 1 m^3 to 10 m^3 at a temperature of 293 K. Use the values $a = 15.7 \times 10^4 \text{ Nm}^4$, $b = 1.07 \times 10^{-2} \text{ m}^3$, and $R = 0.278 \text{ kJ/kg}\cdot\text{K}$.

Ans. 1742.14 kJ

- 3.15 If a gas of volume 6000 cm^3 and at a pressure of 100 kPa is compressed quasistatically according to $pV^{1/2} = \text{constant}$ until the volume becomes 2000 cm^3 , determine the final pressure and the work transfer.

Ans. 900 kPa, 1.2 kJ

- 3.16 The flow energy of $0.124 \text{ m}^3/\text{min}$ of a fluid crossing a boundary to a system is 18 kW. Find the pressure at this point.

Ans. 764 kPa

- 3.17 A milk chilling unit can remove heat from the milk at the rate of 41.87 MJ/h. Heat leaks into the milk from the surroundings at an average rate of 4.187 MJ/h. Find the time required for cooling a batch of 500 kg of milk from 45°C to 5°C . Take the c_p of milk to be $4.187 \text{ kJ/kg}\cdot\text{K}$.

- 3.18 680 kg of fish at 5°C are to be frozen and stored at -12°C . The specific heat of fish above freezing point is 3.182, and below freezing point is $1.717 \text{ kJ/kg}\cdot\text{K}$. The freezing point is -2°C , and the latent heat of fusion is 234.5 kJ/kg . How much heat must be removed to cool the fish, and what per cent of this is latent heat?

Ans. 186.28 MJ, 85.6%

- 3.19 A horizontal cylinder fitted with a sliding piston contains 0.1 m^3 of a gas at a pressure of 1 atm. The piston is restrained by a linear spring. In the initial state, the gas pressure inside the cylinder just balances the atmospheric pressure of 1 atm on the outside of the piston and the spring exerts no force on the piston. The gas is then heated reversibly until its volume and pressure become 0.16 m^3 and 2 atm, respectively. (a) Write the equation for the relation between the pressure and volume of the gas. (b) Calculate the work done by the gas. (c) Of the total work done by the gas, how much is done against the atmosphere? How much is done against the spring?

Ans. (a) $p \text{ (N/m}^2\text{)} = 2.026 \times 10^6 V - 1.013 \times 10^5$

(b) 7,598 J, (c) 5,065 J, 2,533 J

- 3.20 An elastic sphere initially has a diameter of 1 m and contains a gas at a pressure of 1 atm. Due to heat transfer the diameter of the sphere increases to 1.1 m. During the heating process the gas pressure inside the sphere is proportional to the sphere diameter. Calculate the work done by the gas.

Ans. 18.4 kJ

- 3.21 A piston-cylinder device contains 0.05 m^3 of a gas initially at 200 kPa. At this state, a linear spring having a spring constant of 150 kN/m is touching the piston but exerting no force on it. Now heat is transferred to the gas, causing the piston to rise and to compress the spring until the volume inside the cylinder doubles. If the cross-sectional area of the piston is 0.25 m^2 , determine (a) the final pressure

inside the cylinder, (b) the total work done by the gas, and (c) the fraction of this work done against the spring to compress it.

Ans. (a) 320 kPa, (b) 13 kJ, (c) 3 kJ

- 3.22 A piston-cylinder device, whose piston is resting on a set of stops, initially contains 3 kg of air at 200 kPa and 27°C. The mass of the piston is such that a pressure of 400 kPa is required to move it. Heat is now transferred to the air until its volume doubles. Determine the work done by the air and the total heat transferred to the air.

Ans. 516 kJ, 2674 kJ

First Law of Thermodynamics

Energy can be in two forms: (a) energy in *transit*, like heat and work transfer observed at the boundaries of a system, and (b) energy in *storage*, where energy is stored either *macroscopically* by virtue of motion, position or configuration of the system, or *microscopically* in the molecules or atoms constituting the system.

4.1 First Law for a Closed System Undergoing a Cycle

The transfer of heat and the performance of work may both cause the same effect in a system. Heat and work are different forms of the same entity, called energy, which is conserved. Energy which enters a system as heat may leave the system as work, or energy which enters the system as work may leave as heat.

Let us consider a closed system which consists of a known mass of water contained in an adiabatic vessel having a thermometer and a paddle wheel, as shown in Fig. 4.1. Let a certain amount of work W_{1-2} be done upon the system by the paddle wheel. The quantity of work can be measured by the fall of weight which drives the paddle wheel through a pulley. The system was initially at temperature t_1 , the same as that of atmosphere, and after work transfer let the temperature rise to t_2 . The pressure is always 1 atm. The process 1-2 undergone by the system is shown in Fig. 4.2 in generalized thermodynamic coordinates X , Y . Let the insulation now be removed. The system and the surroundings interact by heat transfer till the system returns to the original temperature t_1 , attaining the condition of thermal equilibrium with the atmosphere. The amount of heat transfer Q_{2-1} from the system during this process, 2-1, shown in Fig. 4.2, can be estimated. The system thus executes a cycle, which consists of a definite amount of work input W_{1-2} to the system followed by the transfer of an amount of heat Q_{2-1} from the system. It has been found that this W_{1-2} is always proportional to the heat Q_{2-1} , and the constant of proportionality is called the Joule's equivalent

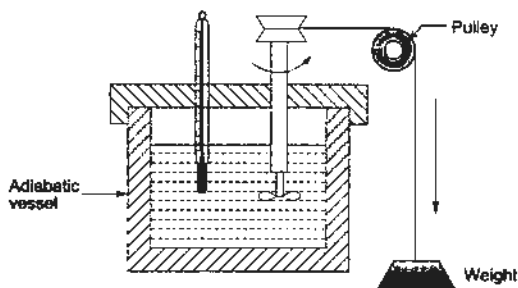


Fig. 4.1 Adiabatic work

or the *mechanical equivalent of heat*. In the simple example given here, there are only two energy transfer quantities as the system performs a thermodynamic cycle. If the cycle involves many more heat and work quantities, the same result will be found. Expressed algebraically,

$$(\Sigma W)_{\text{cycle}} = J (\Sigma Q)_{\text{cycle}} \quad (4.1)$$

where J is the Joule's equivalent. This is also expressed in the form

$$\oint dW = J \oint dQ$$

where the symbol $\oint$ denotes the cyclic integral for the closed path. This is the *first law for a closed system undergoing a cycle*. It is accepted as a *general law of nature*, since no violation of it has ever been demonstrated.

In the S.I. system of units, both heat and work are measured in the derived unit of energy, the Joule. The constant of proportionality, J , is therefore unity ($J = 1 \text{ Nm/J}$).

The first law of thermodynamics owes much to J.P. Joule who, during the period 1840–1849, carried out a series of experiments to investigate the equivalence of work and heat. In one of these experiments, Joule used an apparatus similar to the one shown in Fig. 4.1. Work was transferred to the

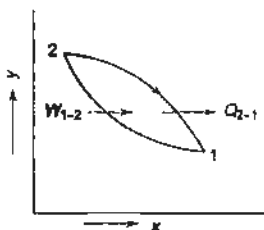


Fig. 4.2 Cycle completed by a system with two energy interactions: adiabatic work transfer W_{1-2} followed by heat transfer Q_{2-1}

measured mass of water by means of a paddle wheel driven by the falling weight. The rise in the temperature of water was recorded. Joule also used mercury as the fluid system, and later a solid system of metal blocks which absorbed work by friction when rubbed against each other. Other experiments involved the supplying of work in an electric current. In every case, he found the same ratio (J) between the amount of work and the quantity of heat that would produce identical effects in the system.

Prior to Joule, heat was considered to be an invisible fluid flowing from a body of higher calorie to a body of lower calorie, and this was known as the *caloric theory of heat*. It was Joule who first established that heat is a form of energy, and thus laid the foundation of the first law of thermodynamics.

4.2 First Law for a Closed System Undergoing a Change of State

The expression $(\Sigma W)_{\text{cycle}} = (\Sigma Q)_{\text{cycle}}$ applies only to systems undergoing cycles, and the algebraic summation of all energy transfer across system boundaries is zero. But if a system undergoes a change of state during which both heat transfer and work transfer are involved, the *net* energy transfer will be stored or accumulated within the system. If Q is the amount of heat transferred to the system and W is the amount of work transferred from the system during the process (Fig. 4.3), the net energy transfer $(Q - W)$ will be stored in the system. Energy in storage is neither heat nor work, and is given the name *internal energy* or simply, the *energy* of the system.

Therefore

$$Q - W = \Delta E$$

where ΔE is the increase in the energy of the system

or

$$Q = \Delta E + W \quad (4.2)$$

Here Q , W , and ΔE are all expressed in the same units (in joules). Energy may be stored by a system in different modes, as explained in Article 4.4.

If there are more energy transfer quantities involved in the process, as shown in Fig. 4.4, the first law gives

$$(Q_2 + Q_3 - Q_1) = \Delta E + (W_2 + W_3 - W_1 - W_4)$$

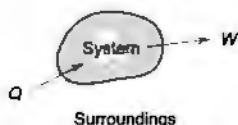


Fig. 4.3 Heat and work interactions of a system with its surroundings in a process

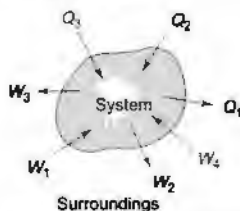


Fig. 4.4 System-surroundings interaction in a process involving many energy fluxes

Energy is thus conserved in the operation. The first law is a particular formulation of the principle of the conservation of energy. Equation (4.2) may also be considered as the definition of energy. This definition does not give an absolute value of energy E , but only the change of energy ΔE for the process. It can, however, be shown that the energy has a definite value at every state of a system and is, therefore, a property of the system.

4.3 Energy—A Property of the System

Consider a system which changes its state from state 1 to state 2 by following the path A , and returns from state 2 to state 1 by following the path B (Fig. 4.5). So the system undergoes a cycle. Writing the first law for path A

$$Q_A = \Delta E_A + W_A \quad (4.3)$$

and for path B

$$Q_B = \Delta E_B + W_B \quad (4.4)$$

The processes A and B together constitute a cycle, for which

$$\begin{aligned} (\Sigma W)_{\text{cycle}} &= (\Sigma Q)_{\text{cycle}} \\ \text{or } W_A + W_B &= Q_A + Q_B \\ \text{or } Q_A - W_A &= W_B - Q_B \end{aligned} \quad (4.5)$$

From equations (4.3), (4.4), and (4.5), it yields

$$\Delta E_A = -\Delta E_B \quad (4.6)$$

Similarly, had the system returned from state 2 to state 1 by following the path C instead of path B

$$\Delta E_A = -\Delta E_C \quad (4.7)$$

From equations (4.6) and (4.7)

$$\Delta E_B = \Delta E_C \quad (4.8)$$

Therefore, it is seen that the change in energy between two states of a system is the same, whatever path the system may follow in undergoing that change of state. If some arbitrary value of energy is assigned to state 2, the value of energy at state 1 is fixed independent of the path the system follows. Therefore, energy has a definite value for every state of the system. Hence, it is a *point function* and a *property of the system*.

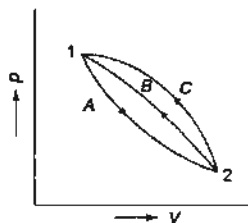


Fig. 4.5 Energy—a property of a system

The energy E is an extensive property. The *specific energy*, $e = E/m$ (J/kg), is an intensive property.

The cyclic integral of any property is zero, because the final state is identical with the initial state. $\oint dE = 0$, $\oint dV = 0$, etc. So for a cycle, the equation (4.2) reduces to equation (4.1).

4.4 Different Forms of Stored Energy

The symbol E refers to the total energy stored in a system. Basically there are two modes in which energy may be stored in a system:

- (a) Macroscopic energy mode
- (b) Microscopic energy mode

The macroscopic energy mode includes the macroscopic kinetic energy and potential energy of a system. Let us consider a fluid element of mass m having the centre of mass velocity $\bar{V}$ (Fig. 4.6). The macroscopic kinetic energy E_K of the fluid element by virtue of its motion is given by

$$E_K = \frac{m \bar{V}^2}{2}$$

If the elevation of the fluid element from an arbitrary datum is z , then the macroscopic potential energy E_p by virtue of its position is given by

$$E_p = mgz$$

The microscopic energy mode refers to the energy stored in the molecular and atomic structure of the system, which is called the *molecular internal energy* or *simply internal energy*, customarily denoted by the symbol U . Matter is composed of molecules. Molecules are in random thermal motion (for a gas) with an average velocity $\bar{v}$, constantly colliding with one another and with the walls (Fig. 4.6). Due to a collision, the molecules may be subjected to rotation as well

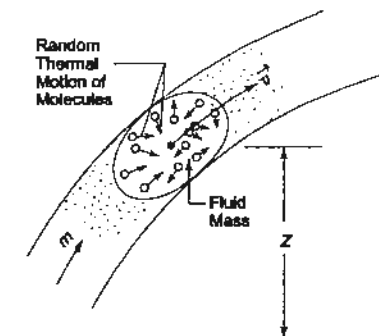


Fig. 4.6 Macroscopic and microscopic energy

as vibration. They can have translational kinetic energy, rotational kinetic energy, vibrational energy, electronic energy, chemical energy and nuclear energy (Fig. 4.7). If ϵ represents the energy of one molecule, then

$$\epsilon = \epsilon_{\text{trans}} + \epsilon_{\text{rot}} + \epsilon_{\text{vib}} + \epsilon_{\text{chem}} + \epsilon_{\text{electronic}} + \epsilon_{\text{nuclear}} \quad (4.9)$$

If N is the total number of molecules in the system, then the total internal energy

$$U = N\epsilon \quad (4.10)$$

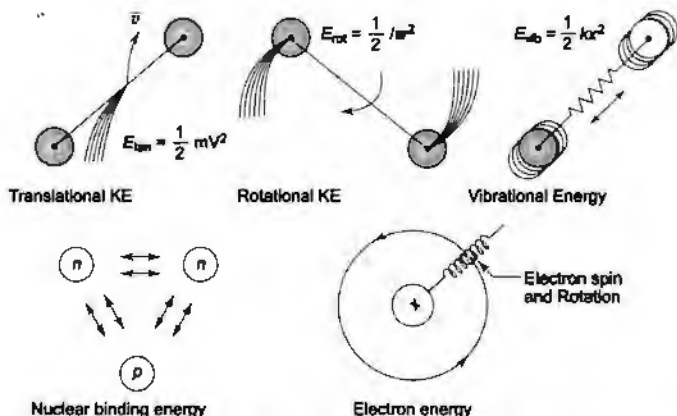


Fig. 4.7 Various components of internal energy stored in a molecule

In an ideal gas there are no intermolecular forces of attraction and repulsion, and the internal energy depends only on temperature. Thus

$$U = f(T) \text{ only} \quad (4.11)$$

for an ideal gas

Other forms of energy which can also be possessed by a system are magnetic energy, electrical energy and surface (tension) energy. In the absence of these forms, the total energy E of a system is given by

$$E = \underbrace{E_K + E_P}_{\text{macro}} + \underbrace{U}_{\text{micro}} \quad (4.12)$$

where E_K , E_P , and U refer to the kinetic, potential and internal energy, respectively. In the absence of motion and gravity

$$E_K = 0, E_P = 0$$

$$E = U$$

and equation (4.2) becomes

$$Q = \Delta U + W \quad (4.13)$$

U is an extensive property of the system. The specific internal energy u is equal to U/m and its unit is J/kg.

In the differential forms, equations (4.2) and (4.13) become

$$\delta Q = dE + \delta W \quad (4.14)$$

$$\delta Q = \delta U + \delta W \quad (4.15)$$

where

$$\delta W = \delta W_{pdV} + \delta W_{\text{shaft}} + \delta W_{\text{electrical}} + \dots,$$

considering the different forms of work transfer which may be present. When only pdV work is present, the equations become

$$\delta Q = dE + pdV \quad (4.16)$$

$$\delta Q = dU + pdV \quad (4.17)$$

or, in the integral form

$$Q = \Delta E + \int pdV \quad (4.18)$$

$$Q = \Delta U + \int pdV \quad (4.19)$$

4.5 Specific Heat at Constant Volume

The specific heat of a substance at constant volume c_v is defined as the rate of change of specific internal energy with respect to temperature when the volume is held constant, i.e.

$$c_v = \left(\frac{\partial u}{\partial T} \right)_v \quad (4.20)$$

For a constant-volume process

$$(\Delta u)_v = \int_{T_1}^{T_2} c_v \cdot dT \quad (4.21)$$

The first law may be written for a closed stationary system composed of a unit mass of a pure substance

$$Q = \Delta u + W$$

or

$$\delta Q = du + \delta W$$

For a process in the absence of work other than pdV work

$$\delta W = p \delta V$$

$\therefore$

$$\delta Q = du + p dv \quad (4.22)$$

When the volume is held constant

$$(Q)_v = (\Delta u)_v$$

$\therefore$

$$(Q)_v = \int_{T_1}^{T_2} c_v dT \quad (4.23)$$

Heat transferred at constant volume increases the internal energy of the system. If the specific heat of a substance is defined in terms of heat transfer, then

$$c_v = \left(\frac{\partial Q}{\partial T} \right)_v$$

Since Q is not a property, this definition does not imply that c_v is a property of a substance. Therefore, this is not the appropriate method of defining the specific heat, although $(dQ)_v = du$.

Since u , T , and v are properties, c_v is a property of the system. The product $mc_v = C_v$ is called the *heat capacity at constant volume* (J/K).

4.6 Enthalpy

The enthalpy of a substance, h , is defined as

$$h = u + pv \quad (4.24)$$

It is an intensive property of a system (kJ/kg).

Internal energy change is equal to the heat transferred in a constant volume process involving no work other than pdV work. From equation (4.22), it is possible to derive an expression for the heat transfer in a constant pressure process involving no work other than pdV work. In such a process in a closed stationary system of unit mass of a pure substance

$$dQ = du + pdv$$

At constant pressure

$$pdv = d(pv)$$

$$\therefore (dQ)_p = du + d(pv)$$

$$\text{or} \quad (dQ)_p = d(u + pv)$$

$$\text{or} \quad (dQ)_p = dh \quad (4.25)$$

where $h = u + pv$ is the *specific enthalpy*, a property of the system.

Heat transferred at constant pressure increases the enthalpy of a system.

For an ideal gas, the enthalpy becomes

$$h = u + RT \quad (4.26)$$

Since the internal energy of an ideal gas depends only on the temperature (Eq. 4.11), the enthalpy of an ideal gas also depends on the temperature only, i.e.

$$h = f(T) \text{ only} \quad (4.27)$$

Total enthalpy $H = mh$

$$\text{Also} \quad H = U + pV$$

$$\text{and} \quad h = H/m \text{ (J/kg)}$$

4.7 Specific Heat at Constant Pressure

The specific heat at constant pressure c_p is defined as the rate of change of enthalpy with respect to temperature when the pressure is held constant

$$c_p = \left(\frac{\partial h}{\partial T} \right)_p \quad (4.28)$$

Since h , T and p are properties, so c_p is a property of the system. Like c_v , c_p should not be defined in terms of heat transfer of constant pressure, although $(dQ)_p = dh$.

For a constant pressure process

$$(\Delta h)_p = \int_{T_1}^{T_2} c_p \cdot dT \quad (4.29)$$

The first law for a closed stationary system of unit mass

$$\delta Q = du + p dv$$

Again

$$h = u + pv$$

$\therefore$

$$dh = du + p dV + v dp$$

$$= dQ + v dp$$

$\therefore$

$$\delta Q = dh - v dp \quad (4.30)$$

$\therefore$

$$(\delta Q)_p = dh$$

or

$$(Q)_p = (\Delta h)_p$$

$\therefore$ From equations (4.19) and (4.20)

$$(Q)_p = \int_{T_1}^{T_2} c_p dT$$

c_p is a property of the system, just like c_v . The *heat capacity at constant pressure* C_p is equal to mc_p (J/K).

4.8 Energy of an Isolated System

An isolated system is one in which there is no interaction of the system with the surroundings. For an isolated system, $\delta Q = 0$, $\delta W = 0$.

The first law gives

$$dE = 0$$

or

$$E = \text{constant}$$

The energy of an isolated system is always constant.

4.9 Perpetual Motion Machine of the First Kind—PMM1

The first law states the general principle of the conservation of energy. *Energy is neither created nor destroyed, but only gets transformed from one form to another.* There can be no machine which would continuously supply mechanical work without some other form of energy disappearing simultaneously (Fig. 4.8). Such a *fictitious machine* is called a *perpetual motion machine of the first kind*, or in brief, PMM1. *A PMM1 is thus impossible.*

The converse of the above statement is also true, i.e., there can be no machine which would continuously consume work without some other form of energy appearing simultaneously (Fig. 4.9).

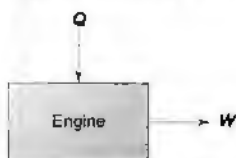


Fig. 4.8 A PMM1

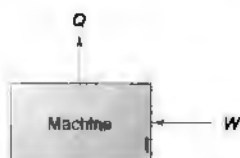


Fig. 4.9 The converse of PMM1

4.10 Limitations of the First Law

The first law deals with the amounts of energy of various forms transferred between the system and its surroundings and with changes in the energy stored in the system. It treats work and heat interactions as equivalent forms of energy in transit and does not indicate the possibility of a spontaneous process proceeding in a certain direction. It is the second law which assigns a quality to different forms of energy, and also indicates the direction of any spontaneous process.

SOLVED EXAMPLES

Example 4.1 A stationary mass of gas is compressed without friction from an initial state of 0.3 m^3 and 0.105 MPa to a final state of 0.15 m^3 and 0.105 MPa , the pressure remaining constant during the process. There is a transfer of 37.6 kJ of heat from the gas during the process. How much does the internal energy of the gas change?

Solution First law for a stationary system in a process gives

$$Q = \Delta U + W$$

$$\text{or} \quad Q_{1-2} = U_2 - U_1 + W_{1-2} \quad (1)$$

Here

$$\begin{aligned} W_{1-2} &= \int_{V_1}^{V_2} p dV = p(V_2 - V_1) \\ &= 0.105 (0.15 - 0.30) \text{ MJ} \\ &= -15.75 \text{ kJ} \\ Q_{1-2} &= -37.6 \text{ kJ} \end{aligned}$$

$\therefore$ Substituting in equation (1)

$$-37.6 \text{ kJ} = U_2 - U_1 - 15.75 \text{ kJ}$$

$$\therefore \quad U_2 - U_1 = -21.85 \text{ kJ}$$

Ans.

The internal energy of the gas decreases by 21.85 kJ in the process.

Example 4.2 When a system is taken from state a to state b , in Fig. Ex. 4.2, along path acb , 84 kJ of heat flow into the system, and the system does 32 kJ of work. (a) How much will the heat that flows into the system along path adb be, if the work done is 10.5 kJ? (b) When the system is returned from b to a along the curved path, the work done on the system is 21 kJ. Does the system absorb or liberate heat, and how much of the heat is absorbed or liberated? (c) If $U_a = 0$ and $U_d = 42$ kJ, find the heat absorbed in the processes ad and db .

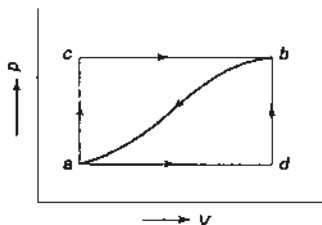


Fig. Ex. 4.2

Solution

$$Q_{acb} = 84 \text{ kJ}$$

$$W_{acb} = 32 \text{ kJ}$$

We have

$$Q_{acb} = U_b - U_a + W_{acb}$$

$$\therefore U_b - U_a = 84 - 32 = 52 \text{ kJ} \quad \text{Ans.}$$

$$\begin{aligned} \text{(a)} \quad Q_{adb} &= U_b - U_a + W_{adb} \\ &= 52 + 10.5 \\ &= 62.5 \text{ kJ} \end{aligned} \quad \text{Ans.}$$

$$\begin{aligned} \text{(b)} \quad Q_{ba} &= U_a - U_b + W_{ba} \\ &= -52 - 21 \\ &= -73 \text{ kJ} \end{aligned} \quad \text{Ans.}$$

The system liberates 73 kJ of heat.

$$\text{(c)} \quad W_{adb} = W_{ad} + W_{db} = W_{ad} = 10.5 \text{ kJ}$$

$$\begin{aligned} \therefore Q_{ad} &= U_d - U_a + W_{ad} \\ &= 42 - 0 + 10.5 = 52.5 \text{ kJ} \end{aligned}$$

$$\text{Now} \quad Q_{adb} = 62.5 \text{ kJ} = Q_{ad} + Q_{db}$$

$$\therefore Q_{db} = 62.5 - 52.5 = 10 \text{ kJ} \quad \text{Ans.}$$

Example 4.3 A piston and cylinder machine contains a fluid system which passes through a complete cycle of four processes. During a cycle, the sum of all heat transfers is -170 kJ. The system completes 100 cycles per min. Complete the

following table showing the method for each item, and compute the net rate of work output in kW.

Process	Q (kJ/min)	W (kJ/min)	ΔE (kJ/min)
a-b	0	2,170	—
b-c	21,000	0	—
c-d	-2,100	—	-36,600
d-a	—	—	—

Solution Process a-b:

$$Q = \Delta E + W$$

$$0 = \Delta E + 2170$$

$\therefore$

$$\Delta E = -2170 \text{ kJ/min}$$

Process b-c:

$$Q = \Delta E + W$$

$$21,000 = \Delta E + 0$$

$\therefore$

$$\Delta E = 21,000 \text{ kJ/min}$$

Process c-d:

$$Q = \Delta E + W$$

$$-2100 = -36,600 + W$$

$\therefore$

$$W = 34,500 \text{ kJ/min}$$

Process d-a:

$$\sum_{\text{cycle}} Q = -170 \text{ kJ}$$

The system completes 100 cycles/min.

$$\therefore Q_{ab} + Q_{bc} + Q_{cd} + Q_{da} = -17,000 \text{ kJ/min}$$

$$0 + 21,000 - 2,100 + Q_{da} = -17,000$$

$\therefore$

$$Q_{da} = -35,900 \text{ kJ/min}$$

Now $\oint dE = 0$, since cyclic integral of any property is zero.

$$\therefore \Delta E_{a-b} + \Delta E_{b-c} + \Delta E_{c-d} + \Delta E_{d-a} = 0$$

$$-2,170 + 21,000 - 36,600 + \Delta E_{d-a} = 0$$

$\therefore$

$$\Delta E_{d-a} = 17,770 \text{ kJ/min}$$

$\therefore$

$$\begin{aligned} W_{d-a} &= Q_{d-a} - \Delta E_{d-a} \\ &= -35,900 - 17,770 \\ &= -53,670 \text{ kJ/min} \end{aligned}$$

The table becomes

Process	Q (kJ/min)	W (kJ/min)	ΔE (kJ/min)
a-b	0	2,170	-2,170
b-c	21,000	0	21,000

c-d	- 2,100	34,500	- 36,600
d-a	- 35,900	- 53,670	17,770

Since

Rate of work output

$$\begin{aligned}\sum_{\text{cycle}} Q &= \sum_{\text{cycle}} W \\ &= -17,000 \text{ kJ/min} \\ &= -283.3 \text{ kW}\end{aligned}$$

Ans.

Example 4.4 The internal energy of a certain substance is given by the following equation

$$u = 3.56 pv + 84$$

where u is given in kJ/kg, p is in kPa, and v is in m^3/kg .

A system composed of 3 kg of this substance expands from an initial pressure of 500 kPa and a volume of 0.22 m^3 to a final pressure 100 kPa in a process in which pressure and volume are related by $pv^{1.2} = \text{constant}$.

- If the expansion is quasi-static, find Q , ΔU , and W for the process.
- In another process the same system expands according to the same pressure-volume relationship as in part (a), and from the same initial state to the same final state as in part (a), but the heat transfer in this case is 30 kJ. Find the work transfer for this process.
- Explain the difference in work transfer in parts (a) and (b).

Solution

(a)

$$u = 3.56 pv + 84$$

$$\Delta u = u_2 - u_1 = 3.56 (p_2 v_2 - p_1 v_1)$$

$\therefore$

$$\Delta U = 3.56 (p_2 V_2 - p_1 V_1)$$

Now

$$p_1 V_1^{1.2} = p_2 V_2^{1.2}$$

$\therefore$

$$V_2 = V_1 \left(\frac{p_1}{p_2} \right)^{1/1.2} = 0.22 \left(\frac{5}{1} \right)^{1/1.2}$$

$$= 0.22 \times 3.83 = 0.845 \text{ m}^3$$

$\therefore$

$$\Delta U = 356 (1 \times 0.845 - 5 \times 0.22) \text{ kJ}$$

$$= -356 \times 0.255 = -91 \text{ kJ}$$

Ans. (a)

For a quasi-static process

$$W = \int p dV = \frac{p_2 V_2 - p_1 V_1}{1 - n}$$

$$= \frac{(1 \times 0.845 - 5 \times 0.22)100}{1 - 1.2} = 127.5 \text{ kJ}$$

$\therefore$

$$Q = \Delta U + W$$

$$= -91 + 127.5 = 36.5 \text{ kJ}$$

Ans. (a)

(b) Here $Q = 30 \text{ kJ}$

Since the end states are the same, ΔU would remain the same as in (a).

$$\begin{aligned}\therefore W &= Q - \Delta U \\ &= 30 - (-91) \\ &= 121 \text{ kJ}\end{aligned}$$

Ans. (b)

(c) The work in (b) is not equal to $\int p dV$ since the process is not quasi-static.

Example 4.5 A fluid is confined in a cylinder by a spring-loaded, frictionless piston so that the pressure in the fluid is a linear function of the volume ($p = a + bV$). The internal energy of the fluid is given by the following equation

$$U = 34 + 3.15 pV$$

where U is in kJ, p in kPa, and V in cubic metre. If the fluid changes from an initial state of 170 kPa, 0.03 m^3 to a final state of 400 kPa, 0.06 m^3 , with no work other than that done on the piston, find the direction and magnitude of the work and heat transfer.

Solution The change in the internal energy of the fluid during the process.

$$\begin{aligned}U_2 - U_1 &= 3.15 (p_2 V_2 - p_1 V_1) \\ &= 315 (4 \times 0.06 - 1.7 \times 0.03) \\ &= 315 \times 0.189 = 59.5 \text{ kJ}\end{aligned}$$

Now

$$\begin{aligned}p &= a + bV \\ 170 &= a + b \times 0.03 \\ 400 &= a + b \times 0.06\end{aligned}$$

From these two equations

$$\begin{aligned}a &= -60 \text{ kN/m}^2 \\ b &= 7667 \text{ kN/m}^5\end{aligned}$$

Work transfer involved during the process

$$\begin{aligned}W_{1-2} &= \int_{V_1}^{V_2} p dV = \int_{V_1}^{V_2} (a + bV) dV \\ &= a(V_2 - V_1) + b \frac{V_2^2 - V_1^2}{2} \\ &= (V_2 - V_1) \left[a + \frac{b}{2} (V_1 + V_2) \right] \\ &= 0.03 \text{ m}^3 \left[-60 \text{ kN/m}^2 + \frac{7667}{2} \frac{\text{kN}}{\text{m}^5} \times 0.09 \text{ m}^3 \right] \\ &= 8.55 \text{ kJ}\end{aligned}$$

Work is done by the system, the magnitude being 8.55 kJ.

∴ Heat transfer involved is given by

$$\begin{aligned} Q_{1-2} &= U_2 - U_1 + W_{1-2} \\ &= 59.5 + 8.55 \\ &= 68.05 \text{ kJ} \end{aligned}$$

68.05 kJ of heat flow into the system during the process.

REVIEW QUESTIONS

- 4.1 State the first law for a closed system undergoing a cycle.
- 4.2 What was the contribution of J.P. Joule in establishing the first law?
- 4.3 What is the caloric theory of heat? Why was it rejected?
- 4.4 Which is the property introduced by the first law?
- 4.5 State the first law for a closed system undergoing a change of state.
- 4.6 Show that energy is a property of a system.
- 4.7 What are the modes in which energy is stored in a system?
- 4.8 Define internal energy. How is energy stored in molecules and atoms?
- 4.9 What is the difference between the standard symbols of E and U ?
- 4.10 What is the difference between heat and internal energy?
- 4.11 Define enthalpy. Why does the enthalpy of an ideal gas depend only on temperature?
- 4.12 Define the specific heats at constant volume and constant pressure.
- 4.13 Why should specific heat not be defined in terms of heat transfer?
- 4.14 Which property of a system increases when heat is transferred: (a) at constant volume, (b) at constant pressure?
- 4.15 What is a PMM1? Why is it impossible?

PROBLEMS

- 4.1 An engine is tested by means of a water brake at 1000 rpm. The measured torque of the engine is 10000 mN and the water consumption of the brake is $0.5 \text{ m}^3/\text{s}$, its inlet temperature being 20°C . Calculate the water temperature at exit, assuming that the whole of the engine power is ultimately transformed into heat which is absorbed by the cooling water.

Ans. 20.5°C

- 4.2 In a cyclic process, heat transfers are $+14.7 \text{ kJ}$, -25.2 kJ , -3.56 kJ and $+31.5 \text{ kJ}$. What is the net work for this cycle process?

Ans. 17.34 kJ

- 4.3 A slow chemical reaction takes place in a fluid at the constant pressure of 0.1 MPa . The fluid is surrounded by a perfect heat insulator during the reaction which begins at state 1 and ends at state 2. The insulation is then removed and 105 kJ of heat flow to the surroundings as the fluid goes to state 3. The following data are observed for the fluid at states 1, 2 and 3.

State	$V(\text{m}^3)$	$T(^\circ\text{C})$
1	0.003	20

2	0.3	370
3	0.06	20

For the fluid system, calculate E_2 and E_3 , if $E_1 = 0$

Ans. $E_2 = -29.7 \text{ kJ}$, $E_3 = -110.7 \text{ kJ}$

- 4.4 During one cycle the working fluid in an engine engages in two work interactions: 15 kJ to the fluid and 44 kJ from the fluid, and three heat interactions, two of which are known: 75 kJ to the fluid and 40 kJ from the fluid. Evaluate the magnitude and direction of the third heat transfer.

Ans. -6 kJ

- 4.5 A domestic refrigerator is loaded with food and the door closed. During a certain period the machine consumes 1 kW h of energy and the internal energy of the system drops by 5000 kJ. Find the net heat transfer for the system.

Ans. -8.6 MJ

- 4.6 1.5 kg of liquid having a constant specific heat of 2.5 kJ/kg K is stirred in a well-insulated chamber causing the temperature to rise by 15°C. Find ΔE and W for the process.

Ans. $\Delta E = 56.25 \text{ kJ}$, $W = -56.25 \text{ kJ}$

- 4.7 The same liquid as in Problem 4.6 is stirred in a conducting chamber. During the process 1.7 kJ of heat are transferred from the liquid to the surroundings, while the temperature of the liquid is rising by 15°C. Find ΔE and W for the process.

Ans. $\Delta E = 56.25 \text{ kJ}$, $W = 57.95 \text{ kJ}$

- 4.8 The properties of a certain fluid are related as follows

$$u = 196 + 0.718 t$$

$$pv = 0.287 (t + 273)$$

where u is the specific internal energy (kJ/kg), t is in °C, p is pressure (kN/m²), and v is specific volume (m³/kg).

For this fluid, find c_v and c_p

Ans. 0.718, 1.005 kJ/kg K

- 4.9 A system composed of 2 kg of the above fluid expands in a frictionless piston and cylinder machine from an initial state of 1 MPa, 100°C to a final temperature of 30°C. If there is no heat transfer, find the net work for the process.

Ans. 100.52 kJ

- 4.10 If all the work in the expansion of Problem 4.9 is done on the moving piston, show that the equation representing the path of the expansion in the pv -plane is given by $pv^{1.4} = \text{constant}$.

- 4.11 A stationary system consisting of 2 kg of the fluid of Problem 4.8 expands in an adiabatic process according to $pv^{1.2} = \text{constant}$. The initial conditions are 1 MPa and 200°C, and the final pressure is 0.1 MPa. Find W and ΔU for the process.

Why is the work transfer not equal to $\int p dV$?

Ans. $W = 216.83$, $\Delta U = -216.83 \text{ kJ}$, $\int p dV = 434.4 \text{ kJ}$

- 4.12 A mixture of gases expands at constant pressure from 1 MPa, 0.03 m³ to 0.06 m³ with 84 kJ positive heat transfer. There is no work other than that done on a piston. Find ΔE for the gaseous mixture.

Ans. 54 kJ

The same mixture expands through the same state path while a stirring device does 21 kJ of work on the system. Find ΔE , W , and Q for the process.

Ans. 54 kJ, -21 kJ , 33 kJ

- 4.13 A mass of 8 kg gas expands within a flexible container so that the p - v relationship is of the form $pv^{1.2} = \text{const}$. The initial pressure is 1000 kPa and the initial volume is 1 m^3 . The final pressure is 5 kPa. If specific internal energy of the gas decreases by 40 kJ/kg , find the heat transfer in magnitude and direction.

Ans. +2615 kJ

- 4.14 A gas of mass 1.5 kg undergoes a quasi-static expansion which follows a relationship $p = a + bV$, where a and b are constants. The initial and final pressures are 1000 kPa and 200 kPa respectively and the corresponding volumes are 0.20 m^3 and 1.20 m^3 . The specific internal energy of the gas is given by the relation

$$u = 1.5 pv - 85 \text{ kJ/kg}$$

where p is in kPa and v is in m^3/kg . Calculate the net heat transfer and the maximum internal energy of the gas attained during expansion.

Ans. 660 kJ, 503.3 kJ

- 4.15 The heat capacity at constant pressure of a certain system is a function of temperature only and may be expressed as

$$C_p = 2.093 + \frac{41.87}{t + 100} \text{ J/K}$$

where t is the temperature of the system in $^{\circ}\text{C}$. The system is heated while it is maintained at a pressure of 1 atmosphere until its volume increases from 2000 cm^3 to 2400 cm^3 and its temperature increases from 0°C to 100°C . (a) Find the magnitude of the heat interaction. (b) How much does the internal energy of the system increase?

Ans. (a) 238.32 J (b) 197.79 J

- 4.16 An imaginary engine receives heat and does work on a slowly moving piston at such rates that the cycle of operation of 1 kg of working fluid can be represented as a circle 10 cm in diameter on a p - v diagram on which $1 \text{ cm} = 300 \text{ kPa}$ and $1 \text{ cm} = 0.1 \text{ m}^3/\text{kg}$. (a) How much work is done by each kg of working fluid for each cycle of operation? (b) The thermal efficiency of an engine is defined as the ratio of work done and heat input in a cycle. If the heat rejected by the engine in a cycle is 1000 kJ per kg of working fluid, what would be its thermal efficiency?

Ans. (a) 2356.19 kJ/kg, (b) 0.702

- 4.17 A gas undergoes a thermodynamic cycle consisting of three processes beginning at an initial state where $p_1 = 1 \text{ bar}$, $V_1 = 1.5 \text{ m}^3$ and $U_1 = 512 \text{ kJ}$. The processes are as follows:

- Process 1-2: Compression with $pV = \text{constant}$ to $p_2 = 2 \text{ bar}$, $U_2 = 690 \text{ kJ}$
- Process 2-3: $W_{23} = 0$, $Q_{23} = -150 \text{ kJ}$, and
- Process 3-1: $W_{31} = +50 \text{ kJ}$. Neglecting KE and PE changes, determine the heat interactions Q_{12} and Q_{31} .

Ans. 74 kJ, 22 kJ

- 4.18 A gas undergoes a thermodynamic cycle consisting of the following processes:
- Process 1-2: Constant pressure $p = 1.4 \text{ bar}$, $V_1 = 0.028 \text{ m}^3$, $W_{12} = 10.5 \text{ kJ}$.
 - Process 2-3: Compression with $pV = \text{constant}$, $U_3 = U_2$.
 - Process 3-1: Constant volume, $U_1 - U_3 = -26.4 \text{ kJ}$. There are no significant changes in KE and PE. (a) Sketch the cycle on a p - V diagram. (b) Calculate the net work for the

cycle in kJ. (c) Calculate the heat transfer for process 1–2 (d) Show that

$$\sum_{\text{cycle}} Q = \sum_{\text{cycle}} W.$$

Ans. (b) – 8.28 kJ, (c) 36.9 kJ

- 4.19 A certain gas of mass 4 kg is contained within a piston cylinder assembly. The gas undergoes a process for which $pV^{1.5} = \text{constant}$. The initial state is given by 3 bar, 0.1 m^3 . The change in internal energy of the gas in the process is $u_2 - u_1 = -4.6 \text{ kJ/kg}$. Find the net heat transfer for the process when the final volume is 0.2 m^3 . Neglect the changes in KE and PE.

Ans. – 0.8 kJ

- 4.20 An electric generator coupled to a windmill produces an average electrical power output of 5 kW. The power is used to charge a storage battery. Heat transfer from the battery to the surroundings occurs at a constant rate of 0.6 kW. Determine the total amount of energy stored in the battery in 8 h of operation.

Ans. $1.27 \times 10^5 \text{ kJ}$

- 4.21 A gas in a piston-cylinder assembly undergoes two processes in series. From state 1 to state 2 there is energy transfer by heat to the gas of 500 kJ, and the gas does work on the piston amounting 800 kJ. The second process, from state 2 to state 3, is a constant pressure compression at 400 kPa, during which there is a heat transfer from the gas amounting 450 kJ. The following data are also known: $U_1 = 2000 \text{ kJ}$ and $U_3 = 3500 \text{ kJ}$. Neglecting changes in KE and PE, calculate the change in volume of the gas during process 2–3.

Ans. – 5.625 m^3

- 4.22 Air is contained in a rigid well-insulated tank with a volume of 0.2 m^3 . The tank is fitted with a paddle wheel which transfers energy to the air at a constant rate of 4 W for 20 min. The initial density of the air is 1.2 kg/m^3 . If no changes in KE or PE occur, determine (a) the specific volume at the final state, (b) the change in specific internal energy of the air.

Ans. (a) $0.833 \text{ m}^3/\text{kg}$, (b) 20 kJ/kg

5

First Law Applied to Flow Processes

5.1 Control Volume

For any system and in any process, the first law can be written as

$$Q = \Delta E + W$$

where E represents all forms of energy stored in the system.

For a pure substance

$$E = E_K + E_P + U$$

where E_K is the K.E., E_P the P.E., and U the residual energy stored in the molecular structure of the substance.

$$Q = \Delta E_K + \Delta E_P + \Delta U + W \quad (5.1)$$

When there is mass transfer across the system boundary, the system is called an open system. Most of the engineering devices are open systems involving the flow of fluids through them.

Equation (5.1) refers to a system having a particular mass of substance, and is free to move from place to place.

Consider a steam turbine (Fig. 5.1) in which steam enters at a high pressure, does work upon the turbine rotor, and then leaves the turbine at low pressure through the exhaust pipe.

If a certain mass of steam is considered as the thermodynamic system, then the energy equation becomes

$$Q = \Delta E_K + \Delta E_P + \Delta U + W$$

and in order to analyze the expansion process in turbine the moving system is to be followed as it travels through the turbine, taking into account the work and heat interactions all the way through. This method of analysis is similar to that of Lagrange in fluid mechanics.

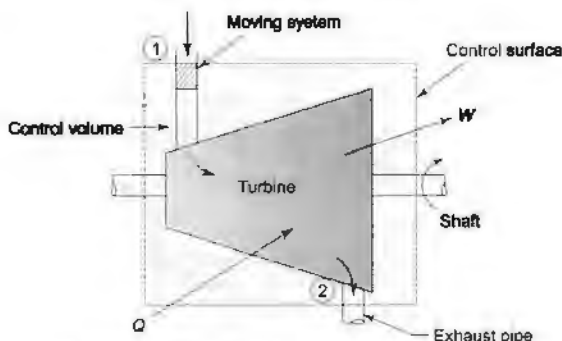


Fig. 5.1 Flow process involving work and heat interactions

Although the system approach is quite valid, there is another approach which is found to be highly convenient. Instead of concentrating attention upon a certain quantity of fluid, which constitutes a moving system in flow process, attention is focussed upon a certain fixed region in space called a *control volume* through which the moving substance flows. This is similar to the analysis of Euler in fluid mechanics.

To distinguish the two concepts, it may be noted that while the system (closed) boundary usually changes shape, position and orientation relative to the observer, the control volume boundary remains fixed and unaltered. Again, while matter usually crosses the control volume boundary, no such flow occurs across the system boundary.

The broken line in Fig. 5.1 represents the surface of the control volume which is known as the *control surface*. This is the same as the system boundary of the open system. The method of analysis is to inspect the control surface and account for all energy quantities transferred through this surface. Since there is mass transfer across the control surface, a mass balance also has to be made. Sections 1 and 2 allow mass transfer to take place, and Q and W are the heat and work interactions respectively.

5.2 Steady Flow Process

As a fluid flows through a certain control volume, its thermodynamic properties may vary along the space coordinates as well as with time. If the rates of flow of mass and energy through the control surface change with time, the mass and energy within the control volume also would change with time.

'Steady flow' means that the rates of flow of mass and energy across the control surface are constant.

In most engineering devices, there is a constant rate of flow of mass and energy through the control surface, and the control volume in course of time attains a steady state. *At the steady state of a system, any thermodynamic property will*

have a fixed value at a particular location, and will not alter with time. Thermodynamic properties may vary along space coordinates, but do not vary with time. 'Steady state' means that the state is steady or invariant with time.

5.3 Mass Balance and Energy Balance in a Simple Steady Flow Process

In Fig. 5.2, a steady flow system has been shown in which one stream of fluid enters and one stream leaves the control volume. There is no accumulation of mass or energy within the control volume, and the properties at any location within the control volume are steady with time. Sections 1.1 and 2.2 indicate, respectively, the entrance and exit of the fluid across the control surface. The following quantities are defined with reference to Fig. 5.2.

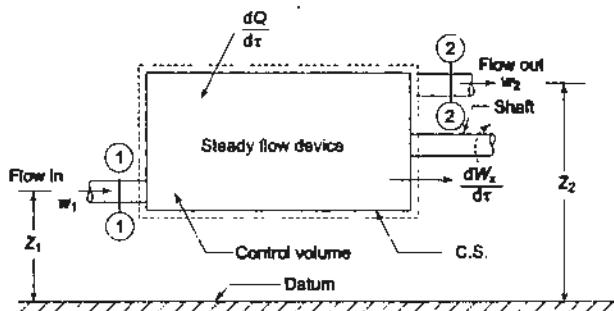


Fig. 5.2 Steady flow Process

A_1, A_2 —cross-section of stream, m^2

w_1, w_2 —mass flow rate, kg/s

p_1, p_2 —pressure, absolute, N/m^2

v_1, v_2 —specific volume, m^3/kg

u_1, u_2 —specific internal energy, J/kg

V_1, V_2 —velocity, m/s

Z_1, Z_2 —elevation above an arbitrary datum, m

$\frac{dQ}{dt}$ —net rate of heat transfer through the control surface, J/s

$\frac{dW_s}{dt}$ —net rate of work transfer through the control surface, J/s

exclusive of work done at sections 1 and 2 in transferring the fluid through the control surface.

t —time, s .

Subscripts 1 and 2 refer to the inlet and exit sections.

5.3.1 Mass Balance

By the conservation of mass, if there is no accumulation of mass within the control volume, the mass flow rate entering must equal the mass flow rate leaving, or

$$\begin{aligned} w_1 &= w_2 \\ \text{or} \quad \frac{A_1 V_1}{v_1} &= \frac{A_2 V_2}{v_2} \end{aligned} \quad (5.2)$$

This equation is known as the *equation of continuity*.

5.3.2 Energy Balance

In a flow process, the work transfer may be of two types: the *external work* and the *flow work*. The external work refers to all the work transfer across the control surface other than that due to normal fluid forces. In engineering thermodynamics the only kinds of external work of importance are *shear* (shaft or stirring) *work* and *electrical work*. In Fig. 5.2 the only external work occurs in the form of shaft work, W_x . The flow work, as discussed in Sec. 3.4, is the displacement work done by the fluid of mass dm_1 at the inlet section 1 and that of mass dm_2 at the exit section 2, which are $(-p_1 v_1 dm_1)$ and $(+p_2 v_2 dm_2)$ respectively. Therefore, the total work transfer is given by

$$W = W_x - p_1 v_1 dm_1 + p_2 v_2 dm_2 \quad (5.3)$$

In the rate form,

$$\begin{aligned} \frac{dW}{d\tau} &= \frac{dW_x}{d\tau} - p_1 v_1 \frac{dm_1}{d\tau} + p_2 v_2 \frac{dm_2}{d\tau} \\ \text{or} \quad \frac{dW}{d\tau} &= \frac{dW_x}{d\tau} - w_1 p_1 v_1 + w_2 p_2 v_2 \end{aligned} \quad (5.4)$$

Since there is no accumulation of energy, by the conservation of energy, the total rate of flow of all energy streams entering the control volume must equal the total rate of flow of all energy streams leaving the control volume. This may be expressed in the following equation.

$$w_1 e_1 + \frac{dQ}{d\tau} = w_2 e_2 + \frac{dW}{d\tau}$$

Substituting for $\frac{dW}{d\tau}$ from Eq. (5.4)

$$w_1 e_1 + \frac{dQ}{d\tau} = w_2 e_2 + \frac{dW_x}{d\tau} - w_1 p_1 v_1 + w_2 p_2 v_2$$

$$w_1 e_1 + w_1 p_1 v_1 + \frac{dQ}{d\tau} = w_2 e_2 + w_2 p_2 v_2 + \frac{dW_x}{d\tau} \quad (5.5)$$

where e_1 and e_2 refer to the energy carried into or out of the control volume with unit mass of fluid.

The specific energy e is given by

$$\begin{aligned} e &= e_k + e_p + u \\ &= \frac{V^2}{2} + Zg + u \end{aligned} \quad (5.6)$$

Substituting the expression for e in Eq. (5.5)

$$\begin{aligned} w_1 \left(\frac{V_1^2}{2} + Z_1g + u_1 \right) + w_1 p_1 v_1 + \frac{dQ}{d\tau} \\ = w_2 \left(\frac{V_2^2}{2} + Z_2g + u_2 \right) + w_2 p_2 v_2 + \frac{dW_x}{d\tau} \end{aligned}$$

$$\begin{aligned} \text{or} \quad w_1 \left(h_1 + \frac{V_1^2}{2} + Z_1g \right) + \frac{dQ}{d\tau} \\ = w_2 \left(h_2 + \frac{V_2^2}{2} + Z_2g \right) + \frac{dW_x}{d\tau} \end{aligned} \quad (5.7)$$

where

$$h = u + pv.$$

And, since

$$w_1 = w_2, \quad \text{let } w = w_1 = w_2 = \frac{dm}{d\tau}$$

Dividing Eq. (5.7) by $\frac{dm}{d\tau}$

$$\begin{aligned} h_1 + \frac{V_1^2}{2} + Z_1g + \frac{dQ}{dm} \\ = h_2 + \frac{V_2^2}{2} + Z_2g + \frac{dW_x}{dm} \end{aligned} \quad (5.8)$$

Equations (5.7) and (5.8) are known as *steady flow energy equations* (S.F.E.E.), for a single stream of fluid entering and a single stream of fluid leaving the control volume. All the terms in Eq. (5.8) represent energy flow per unit mass of fluid (J/kg) whereas all the terms in Eq. (5.7) represent energy flow per unit time (J/kg). The basis of energy flow per unit mass is usually more convenient when only a single stream of fluid enters and leaves a control volume. When more than one fluid stream is involved the basis of energy flow per unit time is more convenient.

Equation (5.8) can be written in the following form,

$$Q - W_x = (h_2 - h_1) + \frac{\bar{V}_2^2 - \bar{V}_1^2}{2} + g(Z_2 - Z_1) \quad (5.9)$$

where Q and W_x refer to energy transfer per unit mass. In the differential form, the SFEE becomes

$$\delta Q - \delta W_x = dh + \bar{V} d\bar{V} + g dZ \quad (5.10)$$

When more than one stream of fluid enters or leaves the control volume (Fig. 5.3), the mass balance and energy balance for steady flow are given below.

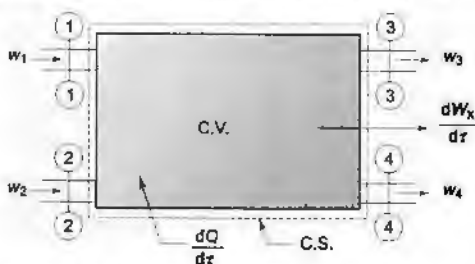


Fig. 5.3 Steady flow process involving two fluid streams at the inlet and exit of the control volume

Mass balance

$$w_1 + w_2 = w_3 + w_4 \quad (5.11)$$

$$\frac{A_1 V_1}{v_1} + \frac{A_2 V_2}{v_2} = \frac{A_3 V_3}{v_3} + \frac{A_4 V_4}{v_4} \quad (5.12)$$

Energy balance

$$\begin{aligned} w_1 \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) + w_2 \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) + \frac{\delta Q}{d\tau} \\ = w_3 \left(h_3 + \frac{V_3^2}{2} + Z_3 g \right) + w_4 \left(h_4 + \frac{V_4^2}{2} + Z_4 g \right) + \frac{\delta W_x}{d\tau} \end{aligned} \quad (5.13)$$

The steady flow energy equation applies to a wide variety of processes like pipe line flows, heat transfer processes, mechanical power generation in engines and turbines, combustion processes, and flows through nozzles and diffusers. In certain problems, some of the terms in steady flow energy equation may be negligible or zero. But it is best to write the full equation first, and then eliminate the terms which are unnecessary.

5.4 Some Examples of Steady Flow Processes

The following examples illustrate the applications of the steady flow energy equation in some of the engineering systems.

5.4.1 Nozzle and Diffusor

A nozzle is a device which increases the velocity or K.E. of a fluid at the expense of its pressure drop, whereas a diffusor increases the pressure of a fluid at the

expense of its K.E. Figure 5.4 shows a nozzle which is insulated. The steady flow energy equation of the control surface gives

$$h_1 + \frac{V_1^2}{2} + Z_1 g + \frac{dQ}{dm} = h_2 + \frac{V_2^2}{2} + Z_2 g + \frac{dW_s}{dm}$$

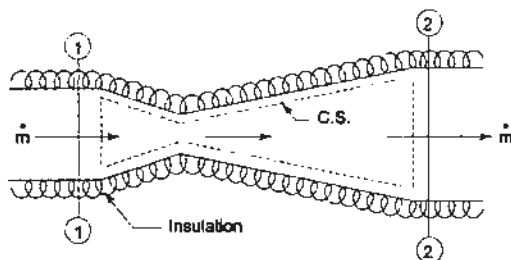


Fig. 5.4 Steady flow process involving one fluid stream at the inlet and at exit of the control volume

Here $\frac{dQ}{dm} = 0$, $\frac{dW_s}{dm} = 0$, and the change in potential energy is zero.

The equation reduces to

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \quad (5.14)$$

The continuity equation gives

$$w = \frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2} \quad (5.15)$$

When the inlet velocity or the 'velocity of approach' V_1 is small compared to the exit velocity V_2 , Eq. (5.14) becomes

$$h_1 = h_2 + \frac{V_2^2}{2}$$

or

$$V_2 = \sqrt{2(h_1 - h_2)} \text{ m/s}$$

where $(h_1 - h_2)$ is in J/kg.

Equations (5.14) and (5.15) hold good for a diffuser as well.

5.4.2 Throttling Device

When a fluid flows through a constricted passage, like a partially opened valve, an orifice, or a porous plug, there is an appreciable drop in pressure, and the flow is said to be throttled. Figure 5.5 shows the process of throttling by a partially opened valve on a fluid flowing in an insulated pipe. In the steady-flow energy Eq. (5.8),

$$\frac{\delta Q}{dm} = 0, \quad \frac{\delta W_x}{dm} = 0$$

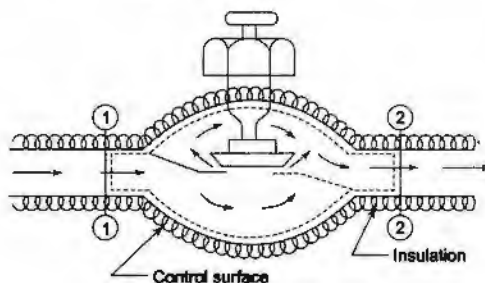


Fig. 5.5 Flow through a valve

and the changes in P.E. are very small and ignored. Thus the S.F.E.E. reduces to

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$$

Often the pipe velocities in throttling are so low that the K.E. terms are also negligible. So

$$h_1 = h_2 \quad (5.16)$$

or the enthalpy of the fluid before throttling is equal to the enthalpy of the fluid after throttling.

5.4.3 Turbine and Compressor

Turbines and engines give positive power output, whereas compressors and pumps require power input.

For a turbine (Fig. 5.6) which is well insulated, the flow velocities are often small, and the K.E. terms can be neglected. The S.F.E.E. then becomes

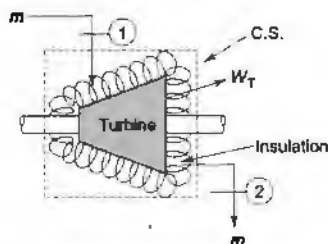


Fig. 5.6 Flow through a turbine

$$h_1 = h_2 + \frac{\delta W_x}{dm}$$

or
$$\frac{W_x}{m} = (h_1 - h_2)$$

It is seen that work is done by the fluid at the expense of its enthalpy.

Similarly, for an adiabatic pump or compressor, work is done upon the fluid and W is negative. So the S.F.E.E. becomes

$$h_1 = h_2 - \frac{W_x}{m}$$

or
$$\frac{W_x}{m} = h_2 - h_1$$

The enthalpy of the fluid increases by the amount of work input.

5.4.4 Heat Exchanger

A heat exchanger is a device in which heat is transferred from one fluid to another. Figure 5.7 shows a steam condenser, where steam condenses outside the tubes and cooling water flows through the tubes. The S.F.E.E. for the C.S. gives

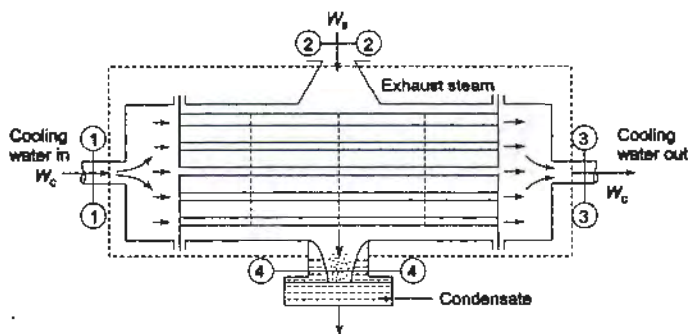


Fig. 5.7 Steam condenser

$$w_c h_1 + w_s h_2 = w_c h_3 + w_s h_4$$

or
$$w_s (h_2 - h_4) = w_c (h_3 - h_1)$$

Here the K.E. and P.E. terms are considered small, there is no external work done, and energy exchange in the form of heat is confined only between the two fluids, i.e., there is no external heat interaction or heat loss.

Figure 5.8 shows a steam desuperheater where the temperature of the superheated steam is reduced by spraying water. If w_1 , w_2 , and w_3 are the mass

flow rates of the injected water, of the steam entering, and of the steam leaving, respectively, and h_1 , h_2 , and h_3 are the corresponding enthalpies, and if K.E. and P.E. terms are neglected as before, the S.F.E.E. becomes

$$w_1 h_1 + w_2 h_2 = w_3 h_3$$

and the mass balance gives

$$w_1 + w_2 = w_3$$

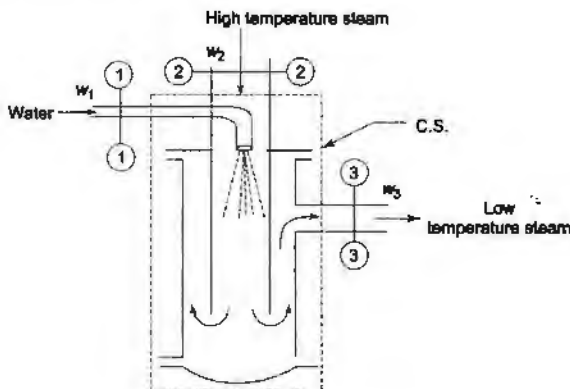


Fig. 5.8 Steam desuperheater

5.5 Comparison of S.F.E.E. with Euler and Bernoulli Equations

The steady flow energy Eq. (5.8) can be written as

$$\frac{dQ}{dm} = (h_2 - h_1) + \frac{\bar{V}_2^2 - \bar{V}_1^2}{2} + (Z_2 - Z_1)g + \frac{dW_x}{dm}$$

In the differential form the S.F.E.E. becomes

$$\delta Q = dh + \bar{V} d\bar{V} + g dZ + dW_x \quad (5.17)$$

where δQ and δW_x refer to unit mass of the substance. Since $h = u + pv$ and

$\delta Q = \delta u + p d\bar{V}$ (for a quasi-static path involving only $p d\bar{V}$ -work), Eq. (5.17) can be written as

$$du + p d\bar{V} = du + p d\bar{V} + \bar{V} dp + \bar{V} d\bar{V} + g dZ + \delta W_x$$

For an inviscid frictionless fluid flowing through a pipe

$$\bar{V} dp + \bar{V} d\bar{V} + g dZ = 0 \quad (5.18)$$

This is the *Euler equation*. If we integrate between two sections 1 and 2 of the pipe

$$\int_1^2 v dp + \int_1^2 v dV + \int_1^2 g dZ = 0$$

For an incompressible fluid, $v = \text{constant}$

$$\therefore v(p_2 - p_1) + \frac{V_2^2}{2} - \frac{V_1^2}{2} + g(Z_2 - Z_1) = 0 \quad (5.19)$$

Since the specific volume v is the reciprocal of the density ρ , we have

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} + Z_1 g = \frac{p_2}{\rho} + \frac{V_2^2}{2} + Z_2 g \quad (5.20)$$

$$\text{or} \quad \frac{p}{\rho} + \frac{V^2}{2} + Zg = \text{constant} \quad (5.21)$$

This is known as the *Bernoulli equation*, which is valid for an inviscid incompressible fluid. It can also be expressed in the following form

$$\Delta \left(p v + \frac{V^2}{2} + gZ \right) = 0 \quad (5.22)$$

where v is constant and $\Delta(\dots)$ means 'increase in...

The S.F.E.E. as given by Eq. (5.18) or Eq. (5.17) can be written with $(u + pv)$ substituted for h , as follows:

$$Q - W_s = \Delta \left(u + pv + \frac{V^2}{2} + gZ \right) \quad (5.23)$$

A comparison of Eqs (5.22) and (5.23) shows that they have several terms in common. However, while the Bernoulli equation is restricted to frictionless incompressible fluids, the S.F.E.E. is not, and is valid for viscous compressible fluids as well. The Bernoulli equation is, therefore, a special limiting case of the more general steady flow energy equation.

5.6 Variable Flow Processes

Many flow processes, such as filling up and evacuating gas cylinders, are not steady. Such processes can be analyzed by the control volume technique. Consider a device through which a fluid is flowing under non-steady state conditions (Fig. 5.9). The rate at which the mass of fluid within the control volume is accumulated as equal to the net rate of mass flow across the control surface, as given below

$$\frac{dm_v}{d\tau} = w_1 - w_2 = \frac{dm_1}{d\tau} - \frac{dm_2}{d\tau} \quad (5.24)$$

where m_v is the mass of fluid within the control volume at any instant.

Over any finite period of time

$$\Delta m_V = \Delta m_1 - \Delta m_2 \quad (5.25)$$

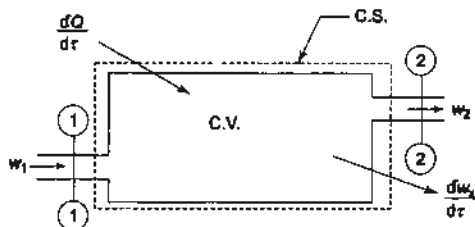


Fig. 5.9 Variable flow process

The rate of accumulation of energy within the control volume is equal to the net rate of energy flow across the control surface. If E_V is the energy of fluid within the control volume at any instant.

Rate of energy increase = Rate of energy inflow – Rate of energy outflow

$$\begin{aligned} \frac{dE_V}{d\tau} &= w_1 \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) + \frac{dQ}{d\tau} \\ &\quad - w_2 \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) - \frac{dW_x}{d\tau} \end{aligned} \quad (5.26)$$

Now

$$E_V = \left(U + \frac{m V^2}{2} + m g Z \right)_V$$

where m is the mass of fluid in the control volume at any instant.

$$\begin{aligned} \therefore \frac{dE_V}{d\tau} &= \frac{d}{d\tau} \left(U + \frac{m V^2}{2} + m g Z \right)_V \\ &= \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) \frac{dm_1}{d\tau} + \frac{dQ}{d\tau} \\ &\quad - \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) \frac{dm_2}{d\tau} - \frac{dW_x}{d\tau} \end{aligned} \quad (5.27)$$

Figure 5.10 shows all these energy flux quantities. For any finite time interval, equation (5.27) becomes

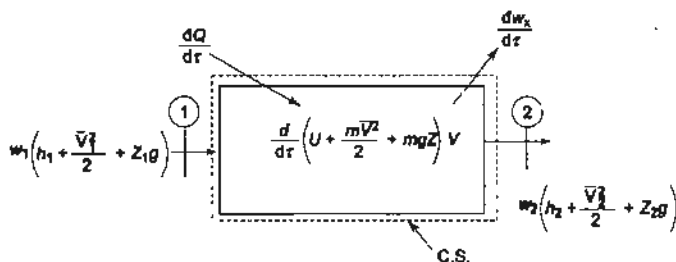


Fig. 5.10 Energy fluxes in an unsteady system

$$\Delta E_V = Q - W_x + \int \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) dm_1 - \int \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) dm_2 \quad (5.28)$$

Equation (5.26) is the *general energy equation*. For steady flow,

$$\frac{dE_V}{d\tau} = 0,$$

and the equation reduces to Eq. (5.7). For a closed system $w_1 = 0$, $w_2 = 0$, then from Eq. (5.26),

$$\frac{dE_V}{d\tau} = \frac{dQ}{d\tau} - \frac{dW_x}{d\tau}$$

or

$$dE_V = dQ - dW_x \text{ or, } dQ = dE + dW_x$$

as obtained earlier.

5.7 Example of a Variable Flow Problem

Variable flow processes may be analyzed either by the system technique or the control volume technique, as illustrated below.

Consider a process in which a gas bottle is filled from a pipeline (Fig. 5.11). In the beginning the bottle contains gas of mass m_1 at state p_1 , t_1 , v_1 , h_1 and u_1 . The valve is opened and gas flows into the bottle till the mass of gas in the bottle is m_2 at state p_2 , t_2 , v_2 , h_2 and u_2 . The supply to the pipeline is very large so that the state of gas in the pipeline is constant at p_p , t_p , v_p , h_p , u_p and V_p .

System Technique Assume an envelope (which is extensible) of gas in the pipeline and the tube which would eventually enter the bottle, as shown in Fig. 5.11.

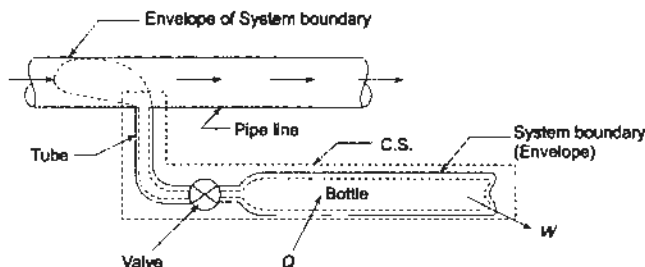


Fig. 5.11 Bottle-filling process

Energy of the gas before filling

$$E_1 = m_1 u_1 + (m_2 - m_1) \left(\frac{V_p^2}{2} + u_p \right)$$

where $(m_2 - m_1)$ is the mass of gas in the pipeline and tube which would enter the bottle.

Energy of the gas after filling

$$E_2 = m_2 u_2$$

$$\Delta E = E_2 - E_1 = m_2 u_2 - \left[m_1 u_1 + (m_2 - m_1) \left(\frac{V_p^2}{2} + u_p \right) \right] \quad (5.29)$$

The P.E. terms are neglected: The gas in the bottle is not in motion, and so the K.E. terms have been omitted.

Now, there is a change in the volume of gas because of the collapse of the envelope to zero volume. Then the work done

$$\begin{aligned} W &= p_p (V_2 - V_1) = p_p [0 - (m_2 - m_1) v_p] \\ &= -(m_2 - m_1) p_p v_p \end{aligned}$$

$\therefore$ Using the first law for the process

$$\begin{aligned} Q &= \Delta E + W \\ &= m_2 u_2 - m_1 u_1 - (m_2 - m_1) \left(\frac{V_p^2}{2} + u_p \right) - (m_2 - m_1) p_p v_p \\ &= m_2 u_2 - m_1 u_1 - (m_2 - m_1) \left(\frac{V_p^2}{2} + h_p \right) \end{aligned} \quad (5.30)$$

which gives the energy balance for the process.

Control Volume Technique Assume a control volume bounded by a control surface, as shown in Fig. 5.11. Applying the energy Eq. (5.27) to this case, the following energy balance may be written on a time rate basis

$$\frac{dE_v}{d\tau} = \frac{dQ}{d\tau} + \left(h_p + \frac{V_p^2}{2} \right) \frac{dm}{d\tau}$$

Since h_p and V_p are constant, the equation is integrated to give for the total process

$$\Delta E_v = Q + \left(h_p + \frac{V_p^2}{2} \right) (m_2 - m_1)$$

Now

$$\Delta E_v = U_2 - U_1 = m_2 u_2 - m_1 u_1$$

$\therefore$

$$Q = m_2 u_2 - m_1 u_1 - \left(h_p + \frac{V_p^2}{2} \right) (m_2 - m_1)$$

This equation is the same as Eq. (5.30).

If $m_1 = 0$, i.e., the bottle is initially evacuated,

$$Q = m_2 u_2 - m_2 \left(h_p + \frac{V_p^2}{2} \right)$$

Again, if $Q = 0$ and $h_p \gg \frac{V_p^2}{2}$,

$$0 = m_2 u_2 - m_2 h_p$$

or,

$$u_2 = h_p = u_p + p_p v_p$$

Thus, flow work ($p_p v_p$) is converted to increase in molecular internal energy ($u_2 - u_p$).

If the gas is assumed ideal,

$$c_v T_2 = c_p T_p$$

or,

$$T_2 = \gamma T_p$$

If $T_p = 27 + 273 = 300$ K, then for air

$$T_2 = 1.4 \times 300 = 420 \text{ K}$$

or,

$$t_2 = 147^\circ\text{C}$$

Therefore, in adiabatically filling a bottle with air at 27°C , the gas temperature rises to 147°C due to the flow work being converted to internal energy increase.

5.8 Discharging and Charging a Tank

Let us consider a tank discharging a fluid into a supply line (Fig. 5.12). Since

$dW_x = 0$ and $dm_{in} = 0$, applying first law to the control volume,

$$dU_v = dQ + \left(h + \frac{V^2}{2} + gz \right)_{out} dm_{out} \quad (5.31)$$

Assuming K.E. and P.E. of the fluid to be small and $dQ = 0$

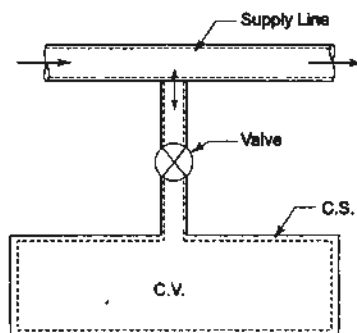


Fig. 5.12 Charging and discharging a tank

$$d(mu) = hdm$$

$$mdu + udm = udm + pv dm$$

$$\frac{dm}{m} = \frac{du}{pv} \quad (5.32)$$

Again

$$V = vm = \text{const.}$$

$$vdm + m dv = 0$$

or

$$\frac{dm}{m} = - \frac{dv}{v} \quad (5.33)$$

From Eqs (5.32) and (5.33),

$$\frac{du}{pv} = - \frac{dv}{v}$$

$$d(u + pv) = 0$$

or

$$\delta Q = 0$$

which shows that the process is adiabatic and quasi-static.

For charging the tank

$$\int (h dm)_m = \Delta U_V = m_2 u_2 - m_1 u_1 \quad (5.34)$$

$$m_p h_p = m_2 u_2 - m_1 u_1$$

where the subscript p refers to the constant state of the fluid in the pipeline. If the tank is initially empty, $m_1 = 0$.

$$m_p h_p = m_2 u_2$$

Since

$$m_p = m_2$$

$$h_p = u_2$$

Enthalpy is converted to internal energy.

If the fluid is an ideal gas, the temperature of the gas in the tank after it is charged is given by

$$\begin{aligned} c_p T_p &= c_v T_2 \\ \text{or} \quad T_2 &= \gamma T_p \end{aligned} \quad (5.35)$$

SOLVED EXAMPLES

Example 5.1 Air flows steadily at the rate of 0.5 kg/s through an air compressor, entering at 7 m/s velocity, 100 kPa pressure, and 0.95 m³/kg volume, and leaving at 5 m/s, 700 kPa, and 0.19 m³/kg. The internal energy of the air leaving is 90 kJ/kg greater than that of the air entering. Cooling water in the compressor jackets absorbs heat from the air at the rate of 58 kW. (a) Compute the rate of shaft work input to the air in kW. (b) Find the ratio of the inlet pipe diameter to outlet pipe diameter.

Solution Figure Ex. 5.1 shows the details of the problem.

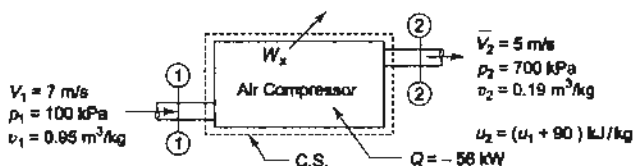


Fig. Ex. 5.1

(a) Writing the steady flow energy equation, we have

$$\begin{aligned} w \left(u_1 + p_1 v_1 + \frac{V_1^2}{2} + Z_1 g \right) + \frac{dQ}{d\tau} \\ = w \left(u_2 + p_2 v_2 + \frac{V_2^2}{2} + Z_2 g \right) + \frac{dW_x}{d\tau} \\ \therefore \frac{dW_x}{d\tau} = -w \left[(u_2 - u_1) + (p_2 v_2 - p_1 v_1) + \frac{V_2^2 - V_1^2}{2} + (Z_2 - Z_1)g \right] + \frac{dQ}{d\tau} \\ \therefore \frac{dW_x}{d\tau} = -0.5 \frac{\text{kg}}{\text{s}} \left[90 \frac{\text{kJ}}{\text{kg}} + (7 \times 0.19 - 1 \times 0.95) 100 \frac{\text{kJ}}{\text{kg}} \right. \\ \left. + \frac{(5^2 - 7^2) \times 10^{-3}}{2} \frac{\text{kJ}}{\text{kg}} + 0 \right] - 58 \text{ kW} \\ = -0.5 [90 + 38 - 0.012] \text{ kJ/s} - 58 \text{ kW} \\ = -122 \text{ kW} \end{aligned}$$

Ans. (a)

Rate of work input is 122 kW.

(b) From mass balance, we have

$$w = \frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2}$$

$$\therefore \frac{A_1}{A_2} = \frac{v_1}{v_2} \cdot \frac{V_2}{V_1} = \frac{0.95}{0.19} \times \frac{5}{7} = 3.57$$

$$\therefore \frac{d_1}{d_2} = \sqrt{3.57} = 1.89 \quad \text{Ans. (b)}$$

Example 5.2 In a steady flow apparatus, 135 kJ of work is done by each kg of fluid. The specific volume of the fluid, pressure, and velocity at the inlet are $0.37 \text{ m}^3/\text{kg}$, 600 kPa, and 16 m/s. The inlet is 32 m above the floor, and the discharge pipe is at floor level. The discharge conditions are $0.62 \text{ m}^3/\text{kg}$, 100 kPa, and 270 m/s. The total heat loss between the inlet and discharge is 9 kJ/kg of fluid. In flowing through this apparatus, does the specific internal energy increase or decrease, and by how much?

Solution Writing the steady flow energy equation for the control volume, as shown in Fig. Ex. 5.2.

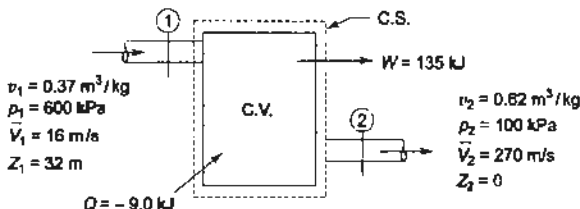


Fig. Ex. 5.2

$$u_1 + p_1 v_1 + \frac{V_1^2}{2} + Z_1 g + \frac{dQ}{dm} = u_2 + p_2 v_2 + \frac{V_2^2}{2} + Z_2 g + \frac{dW_s}{dm}$$

$$\therefore u_1 - u_2 = (p_2 v_2 - p_1 v_1) + \frac{V_2^2 - V_1^2}{2} + (Z_2 - Z_1)g + \frac{dW_s}{dm} - \frac{dQ}{dm}$$

$$= (1 \times 0.62 - 6 \times 0.37) \times 10^2 + \frac{(270^2 - 16^2) \times 10^{-3}}{2}$$

$$+ (-32 \times 9.81 \times 10^{-3}) + 135 - (-9.0)$$

$$= -160 + 36.45 - 0.314 + 135 + 9$$

$$= 20.136 \text{ kJ/kg}$$

Specific internal energy decreases by 20.136 kJ.

Example 5.3 In a steam power station, steam flows steadily through a 0.2 m diameter pipeline from the boiler to the turbine. At the boiler end, the steam conditions are found to be: $p = 4 \text{ MPa}$, $t = 400^\circ\text{C}$, $h = 3213.6 \text{ kJ/kg}$, and $v = 0.073 \text{ m}^3/\text{kg}$. At the turbine end, the conditions are found to be: $p = 3.5 \text{ MPa}$, $t = 392^\circ\text{C}$, $h = 3202.6 \text{ kJ/kg}$, and $v = 0.084 \text{ m}^3/\text{kg}$. There is a heat loss of 8.5 kJ/kg from the pipeline. Calculate the steam flow rate.

Solution Writing the steady flow energy equation for the control volume as shown in Fig. Ex. 5.3

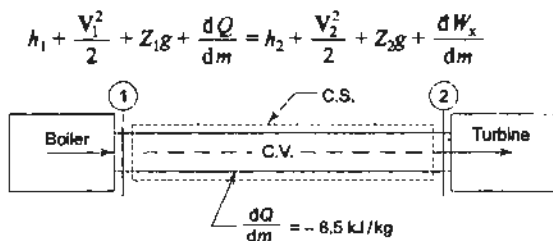


Fig. Ex. 5.3

Here, there is no change in datum, so change in potential energy will be zero.

Now
$$\frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2}$$

$\therefore V_2 = \frac{A_1 V_1}{v_1} \cdot \frac{v_2}{A_2} = \frac{v_2}{v_1} \cdot V_1 = \frac{0.084}{0.073} V_1 = 1.15 V_1$

and
$$\frac{dW_x}{dm} = 0$$

$$h_1 + \frac{V_1^2}{2} + \frac{dQ}{dm} = h_2 + \frac{V_2^2}{2}$$

$$\begin{aligned} \therefore \frac{(V_2^2 - V_1^2) \times 10^{-3}}{2} &= h_1 - h_2 + \frac{dQ}{dm} \\ &= 3213.6 - 3202.6 + (-8.5) \\ &= 2.5 \text{ kJ/kg} \end{aligned}$$

$$\begin{aligned} V_1^2 (1.15^2 - 1^2) &= 5 \times 10^3 \\ V_1^2 &= 15,650 \text{ m}^2/\text{s}^2 \end{aligned}$$

$$\therefore V_1 = 125.1 \text{ m/s}$$

$$\begin{aligned} \therefore \text{Mass flow rate } w &= \frac{A_1 V_1}{v_1} = \frac{\frac{\pi}{4} \times (0.2)^2 \text{ m}^2 \times 125.1 \text{ m/s}}{0.073 \text{ m}^3/\text{kg}} \\ &= 53.8 \text{ kg/s} \end{aligned}$$

Ans.

Example 5.4 A certain water heater operates under steady flow conditions receiving 4.2 kg/s of water at 75°C temperature, enthalpy 313.93 kJ/kg. The water is heated by mixing with steam which is supplied to the heater at temperature 100.2°C and enthalpy 2676 kJ/kg. The mixture leaves the heater as liquid water at temperature 100°C and enthalpy 419 kJ/kg. How much steam must be supplied to the heater per hour?

Solution By mass balance across the control surface (Fig. Ex. 5.4)

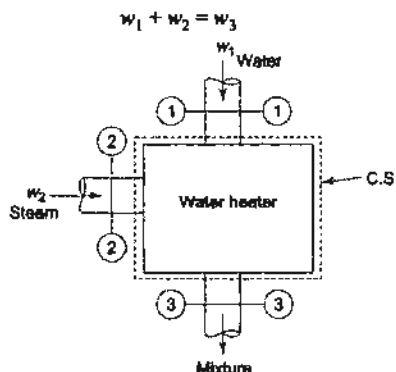


Fig. Ex. 5.4

By energy balance

$$w_1 \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) + \frac{dQ}{dt} + w_2 \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) \\ = w_3 \left(h_3 + \frac{V_3^2}{2} + Z_3 g \right) + \frac{dW_s}{dt}$$

By the nature of the process, there is no shaft work. Potential and kinetic energy terms are assumed to balance zero. The heater is assumed to be insulated. So the steady flow energy equation reduces to

$$w_1 h_1 + w_2 h_2 = w_3 h_3$$

$$4.2 \times 313.93 + w_2 \times 2676 = (4.2 + w_2) 419$$

$$\therefore w_2 = 0.196 \text{ kg/s}$$

$$= 705 \text{ kg/h}$$

Ans.

Example 5.5 Air at a temperature of 15°C passes through a heat exchanger at a velocity of 30 m/s where its temperature is raised to 800°C. It then enters a turbine with the same velocity of 30 m/s and expands until the temperature falls to 650°C. On leaving the turbine, the air is taken at a velocity of 60 m/s to a nozzle

where it expands until the temperature has fallen to 500°C. If the air flow rate is 2 kg/s, calculate (a) the rate of heat transfer to the air in the heat exchanger, (b) the power output from the turbine assuming no heat loss, and (c) the velocity at exit from the nozzle, assuming no heat loss. Take the enthalpy of air as $h = c_p t$, where c_p is the specific heat equal to 1.005 kJ/kg K and t the temperature.

Solution As shown in Fig. Ex. 5.5, writing the S.F.E.E. for the heat exchanger and eliminating the terms not relevant,

$$w \left(h_1 + \frac{V_1^2}{2} + Z_1 g \right) + Q_{1-2} = w \left(h_2 + \frac{V_2^2}{2} + Z_2 g \right) + W_{1-2}$$

$$\therefore wh_1 + Q_{1-2} = wh_2$$

$$\begin{aligned} Q_{1-2} &= w(h_2 - h_1) = w c_p (t_2 - t_1) \\ &= 2 \times 1.005 (800 - 15) \\ &= 2.01 \times 785 \\ &= 1580 \text{ kJ/s} \end{aligned}$$

Ans. (a).

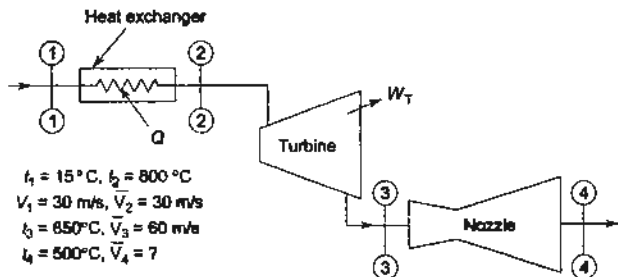


Fig. Ex. 5.5

Energy equation for the turbine gives

$$w \left(\frac{V_2^2}{2} + h_2 \right) = wh_3 + w \frac{V_3^2}{2} + W_T$$

$$\frac{V_2^2 - V_3^2}{2} + (h_2 - h_3) = W_T/w$$

$$\frac{(30^2 - 60^2) \times 10^{-3}}{2} + 1.005 (800 - 650) = W_T/w$$

$$\therefore \frac{W_T}{w} = -1.35 + 150.75$$

$$= 149.4 \text{ kJ/kg}$$

$$\begin{aligned} \therefore W_T &= 149.4 \times 2 \text{ kJ/s} \\ &= 298.8 \text{ kW} \end{aligned}$$

Ans. (b)

Writing the energy equation for the nozzle

$$\begin{aligned}\frac{V_3^2}{2} + h_3 &= \frac{V_4^2}{2} + h_4 \\ \frac{V_4^2 - V_3^2}{2} &= c_p(t_3 - t_4) \\ V_4^2 - V_3^2 &= 1.005 (650 - 500) \times 2 \times 10^3 \\ &= 301.50 \times 10^3 \text{ m}^2/\text{s}^2 \\ V_4^2 &= 30.15 \times 10^4 + 0.36 \times 10^4 \\ &= 30.51 \times 10^4 \text{ m}^2/\text{s}^2\end{aligned}$$

∴ Velocity at exit from the nozzle

$$V_4 = 554 \text{ m/s}$$

Ans. (c)

Example 5.6 The air speed of a turbojet engine in flight is 270 m/s. Ambient air temperature is -15°C . Gas temperature at outlet of nozzle is 600°C . Corresponding enthalpy values for air and gas are respectively 260 and 912 kJ/kg. Fuel-air ratio is 0.0190. Chemical energy of the fuel is 44.5 MJ/kg. Owing to incomplete combustion 5% of the chemical energy is not released in the reaction. Heat loss from the engine is 21 kJ/kg of air. Calculate the velocity of the exhaust jet.

Solution Energy equation for the turbojet engine (Fig. Ex. 5.6) gives

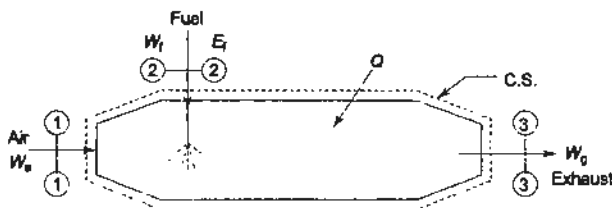


Fig. Ex. 5.6

$$\begin{aligned}w_a \left(h_a + \frac{V_a^2}{2} \right) + w_f E_f + Q &= w_g \left(h_g + \frac{V_g^2}{2} + E_g \right) \\ \left(260 + \frac{270^2 \times 10^{-3}}{2} \right) + 0.0190 \times 44500 - 21 \\ &= 1.0190 \left(912 + \frac{V_g^2 \times 10^{-3}}{2} + 0.05 \frac{0.019}{1.019} \times 44500 \right) \\ 260 + 36.5 + 845 - 21 &= 1.019 \left(912 + \frac{V_g^2 \times 10^{-3}}{2} + 42 \right)\end{aligned}$$

$$\therefore \frac{V_g^2}{2} = 156 \times 10^3 \text{ m}^2/\text{s}^2$$

$$V_g = \sqrt{31.2} \times 100 \text{ m/s}$$

Velocity of exhaust gas, $V_g = 560 \text{ m/s}$

Ans.

Example 5.7 In a reciprocating engine, the mass of gas occupying the clearance volume is m_c kg at state p_1 , u_1 , v_1 and h_1 . By opening the inlet valve, m_f kg of gas is taken into the cylinder, and at the conclusion of the intake process the state of the gas is given by p_2 , u_2 , v_2 , h_2 . The state of the gas in the supply pipe is constant and is given by p_p , u_p , v_p , h_p , V_p . How much heat is transferred between the gas and the cylinder walls during the intake process?

Solution Let us consider the control volume as shown in Fig. Ex. 5.7. Writing the energy balance on a time rate basis

$$\frac{dE_v}{d\tau} = \frac{dQ}{d\tau} - \frac{dW}{d\tau} + \left(h_p + \frac{V_p^2}{2} \right) \frac{dm}{d\tau}$$

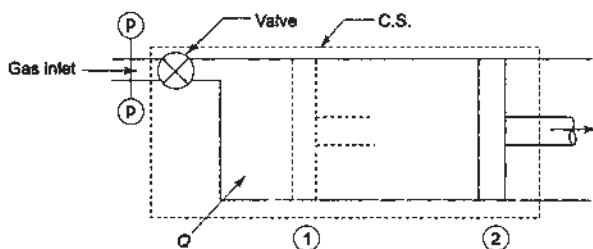


Fig. Ex. 5.7

With h_p and V_p being constant, the above equation can be integrated to give for the total process

$$\Delta E_v = Q - W + \left(h_p + \frac{V_p^2}{2} \right) m_f$$

Now

$$\Delta E_v = U_2 - U_1 = (m_c + m_f) u_2 - m_c u_1$$

$$\therefore Q = (m_c + m_f) u_2 - m_c u_1 - m_f \left(h_p + \frac{V_p^2}{2} \right) + W \quad \text{Ans.}$$

Example 5.8 The internal energy of air is given by

$$u = u_0 + 0.718t$$

where u is in kJ/kg, u_0 is any arbitrary value of u at 0°C , kJ/kg, and t is the temperature in $^\circ\text{C}$. Also for air, $p_v = 0.287 (t + 273)$, where p is in kPa and v is in m^3/kg .

A mass of air is stirred by a paddle wheel in an insulated constant volume tank. The velocities due to stirring make a negligible contribution to the internal energy of the air. Air flows out through a small valve in the tank at a rate controlled to keep the temperature in the tank constant. At a certain instant the conditions are as follows: tank volume 0.12 m^3 , pressure 1 MPa, temperature 150°C , and power to paddle wheel 0.1 kW . Find the rate of flow of air out of the tank at this instant.

Solution Writing the energy balance for the control volume as shown in Fig. Ex. 5.8

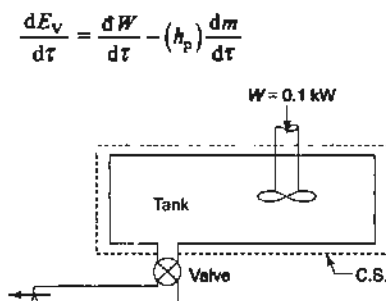


Fig. Ex. 5.8

Since there is no change in internal energy of air in the tank,

$$h_p \frac{dm}{d\tau} = \frac{\delta W}{d\tau}$$

where

$$h_p = u + pv.$$

Let

$$u = 0 \text{ at } t = 0 \text{ K} = -273^\circ\text{C}$$

$$u = u_0 + 0.718 t$$

$$0 = u_0 + 0.718 (-273)$$

$$u_0 = 0.718 \times 273 \text{ kJ/kg}$$

At $^\circ\text{C}$

$$u = 0.718 \times 273 + 0.718 t$$

$$= 0.718 \times (t + 273) \text{ kJ/kg}$$

$$h_p = 0.718 (t + 273) + 0.287 (t + 273)$$

or

$$h_p = 1.005 (t + 273)$$

At 150°C

$$h_p = 1.005 \times 423$$

$$= 425 \text{ kJ/kg}$$

$\therefore$

$$\frac{dm}{d\tau} = \frac{1}{h_p} \frac{\delta W}{d\tau}$$

$$\begin{aligned}
 &= \frac{0.1 \text{ kJ/s}}{425 \text{ kJ/kg}} = 0.236 \times 10^{-3} \text{ kg/s} \\
 &= 0.845 \text{ kg/h}
 \end{aligned}$$

This is the rate at which air flows out of the tank.

Example 5.9 A well-insulated vessel of volume V contains a gas at pressure p_0 and temperature t_0 . The gas from a main at a uniform temperature t_1 is pumped into the vessel and the inflow rate decreases exponentially with time according to $\dot{m} = \dot{m}_0 e^{-a\tau}$, where a is a constant. Determine the pressure and temperature of the gas in the vessel as a function of time. Neglect the K.E. of the gas entering the vessel and assume that the gas follows the relation

$$pv = RT, \text{ where } T = t + 273$$

and its specific heats are constant.

- If the vessel was initially evacuated, show that the temperature inside the vessel is independent of time.
- Determine the charging time when the pressure inside the vessel reaches that of the main.

Solution Since the vessel is well-insulated, $\dot{Q} = 0$ and there is no external work transfer, $\dot{W} = 0$. Therefore,

$$\frac{dE_v}{d\tau} = h_1 \frac{dm}{d\tau} = h_1 \dot{m}_0 e^{-a\tau}$$

where h_1 is the enthalpy of the gas in the main.

On integration,

$$E = E_0 + \frac{h_1 \dot{m}_0}{a} (1 - e^{-a\tau})$$

where E_0 is the initial energy of the vessel at the beginning of the charging process, i.e. $E = E_0$ at $\tau = 0$. Neglecting K.E. and P.E. changes, by energy balance

$$Mu = M_0 u_0 + \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) (u_1 + p_1 v_1) \quad (1)$$

Again,

$$\frac{dm}{d\tau} = \dot{m}_0 e^{-a\tau}$$

On integration,

$$M = M_0 + \frac{\dot{m}_0 (1 - e^{-a\tau})}{a} \quad (2)$$

where M_0 is the initial mass of the gas. Eliminating M from Eqs (1) and (2),

$$\begin{aligned}
 &\left\{ M_0 + \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) \right\} u - M_0 u_0 \\
 &= \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) (u_1 + RT_1)
 \end{aligned}$$

$$M_0 c_v (T - T_0) = \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) \{c_v(T_1 - T) + RT_1\}$$

$$T = \frac{M_0 c_v T_0 + \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) c_p T_1}{\left\{ M_0 + \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) \right\} c_v}$$

$$p = \frac{MRT}{V} = \frac{R}{V c_v} \left\{ M_0 c_v T_0 + \frac{\dot{m}_0}{a} (1 - e^{-a\tau}) c_p T_1 \right\}$$

$$= p_0 + \frac{\dot{m}_0 R}{a V} (1 - e^{-a\tau}) \gamma T_1$$

The above two equations show the temperature and pressure of the gas in the vessel as functions of time.

- (i) If $M_0 = 0$, $T = \gamma T_1$, i.e., the temperature inside the vessel becomes independent of time and is equal to γT_1 throughout the charging process.
- (ii) The charging process will stop when pressure inside the vessel reaches that of the main. The charging time can be found by setting $p = p_1$ in the pressure relation

$$p_1 - p_0 = \frac{\dot{m}_0 R \gamma T_1}{a V} - \frac{\dot{m}_0 R}{a V} e^{-a\tau} \gamma T_1$$

By rearrangement,

$$e^{a\tau} = \frac{\dot{m}_0 R \gamma T_1 / (a V)}{\left(\frac{\dot{m}_0 R \gamma T_1}{a V} \right) - (p_1 - p_0)}$$

$$\tau = -\frac{1}{a} \ln \left[1 - (p_1 - p_0) \frac{a V}{\dot{m}_0 R \gamma T_1} \right]$$

REVIEW QUESTIONS

- 5.1 Explain the system approach and the control volume approach in the analysis of a flow process.
- 5.2 What is a steady flow process? What is steady state?
- 5.3 Write the steady flow energy equation for a single stream entering and a single stream leaving a control volume and explain the various terms in it.
- 5.4 Give the differential form of the S.F.E.E.
- 5.5 Under what conditions does the S.F.E.E. reduce to Euler's equation?
- 5.6 How does Bernoulli's equation compare with S.F.E.E.?
- 5.7 What will be the velocity of a fluid leaving a nozzle, if the velocity of approach is very small?
- 5.8 Show that the enthalpy of a fluid before throttling is equal to that after throttling.
- 5.9 Write the general energy equation for a variable flow process.
- 5.10 What is the system technique in a bottle-filling process?
- 5.11 Explain the control volume technique in a variable flow process.

PROBLEMS

- 5.1 A blower handles 1 kg/s of air at 20°C and consumes a power of 15 kW. The inlet and outlet velocities of air are 100 m/s and 150 m/s respectively. Find the exit air temperature, assuming adiabatic conditions. Take c_p of air as 1.005 kJ/kg·K.

Ans. 28.38°C

- 5.2 A turbine operates under steady flow conditions, receiving steam at the following state: pressure 1.2 MPa, temperature 188°C, enthalpy 2785 kJ/kg, velocity 33.3 m/s and elevation 3 m. The steam leaves the turbine at the following state: pressure 20 kPa, enthalpy 2512 kJ/kg, velocity 100 m/s, and elevation 0 m. Heat is lost to the surroundings at the rate of 0.29 kJ/s. If the rate of steam flow through the turbine is 0.42 kg/s, what is the power output of the turbine in kW?

Ans. 112.51 kW

- 5.3 A nozzle is a device for increasing the velocity of a steadily flowing stream. At the inlet to a certain nozzle, the enthalpy of the fluid passing is 3000 kJ/kg and the velocity is 60 m/s. At the discharge end, the enthalpy is 2762 kJ/kg. The nozzle is horizontal and there is negligible heat loss from it. (a) Find the velocity at exit from the nozzle. (b) If the inlet area is 0.1 m² and the specific volume at inlet is 0.187 m³/kg, find the mass flow rate. (c) If the specific volume at the nozzle exit is 0.498 m³/kg, find the exit area of the nozzle.

Ans. (a) 692.5 m/s, (b) 32.08 kg/s (c) 0.023 m²

- 5.4 In an oil cooler, oil flows steadily through a bundle of metal tubes submerged in a steady stream of cooling water. Under steady flow conditions, the oil enters at 90°C and leaves at 30°C, while the water enters at 25°C and leaves at 70°C. The enthalpy of oil at t°C is given by

$$h = 1.68 t + 10.5 \times 10^{-4} t^2 \text{ kJ/kg}$$

What is the cooling water flow required for cooling 2.78 kg/s of oil?

Ans. 1.473 kg/s

- 5.5 A thermoelectric generator consists of a series of semiconductor elements (Fig. P. 5.5), heated on one side and cooled on the other. Electric current flow is produced as a result of energy transfer as heat. In a particular experiment the

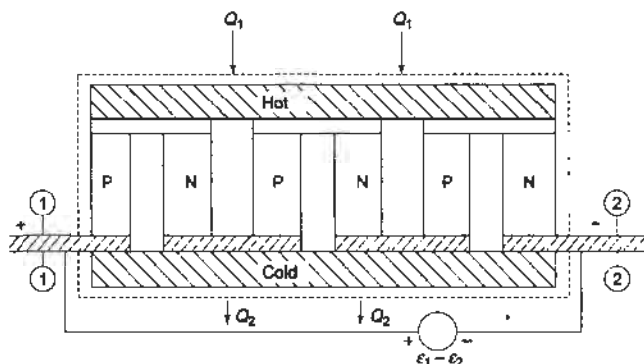


Fig. P.5.5

current was measured to be 0.5 amp and the electrostatic potential at (1) was 0.8 volt above that at (2). Energy transfer as heat to the hot side of the generator was taking place at a rate of 5.5 watts. Determine the rate of energy transfer as heat from the cold side and the energy conversion efficiency.

Ans. $\dot{Q}_2 = 5.1$ watts, $\eta = 0.073$

- 5.6 A turbocompressor delivers $2.33 \text{ m}^3/\text{s}$ of air at 0.276 MPa , 43°C which is heated at this pressure to 430°C and finally expanded in a turbine which delivers 1860 kW . During the expansion, there is a heat transfer of 0.09 MJ/s to the surroundings. Calculate the turbine exhaust temperature if changes in kinetic and potential energy are negligible. Take $c_p = 1.005 \text{ kJ/kgK}$

Ans. 157°C

- 5.7 A reciprocating air compressor takes in $2 \text{ m}^3/\text{min}$ at 0.11 MPa , 20°C which it delivers at 1.5 MPa , 111°C to an aftercooler where the air is cooled at constant pressure to 25°C . The power absorbed by the compressor is 4.15 kW . Determine the heat transfer in (a) the compressor, and (b) the cooler. State your assumptions.

Ans. -0.17 kJ/s , -3.76 kJ/s

- 5.8 In a water cooling tower air enters at a height of 1 m above the ground level and leaves at a height of 7 m . The inlet and outlet velocities are 20 m/s and 30 m/s respectively. Water enters at a height of 8 m and leaves at a height of 0.8 m . The velocity of water at entry and exit are 3 m/s and 1 m/s respectively. Water temperatures are 80°C and 50°C at the entry and exit respectively. Air temperatures are 30°C and 70°C at the entry and exit respectively. The cooling tower is well insulated and a fan of 2.25 kW drives the air through the cooler. Find the amount of air per second required for 1 kg/s of water flow. The values of c_p of air and water are 1.005 and 4.187 kJ/kg K respectively.

Ans. 3.16 kg

- 5.9 Air at 101.325 kPa , 20°C is taken into a gas turbine power plant at a velocity of 140 m/s through an opening of 0.15 m^2 cross-sectional area. The air is compressed heated, expanded through a turbine, and exhausted at 0.18 MPa , 150°C through an opening of 0.10 m^2 cross-sectional area. The power output is 375 kW . Calculate the net amount of heat added to the air in kJ/kg . Assume that air obeys the law $p v = 0.287 (t + 273)$ where p is the pressure in kPa , v is the specific volume in m^3/kg , and t is the temperature in $^\circ\text{C}$. Take $c_p = 1.005 \text{ kJ/kg K}$.

Ans. 150.23 kJ/kg

- 5.10 A gas flows steadily through a rotary compressor. The gas enters the compressor at a temperature of 16°C , a pressure of 100 kPa , and an enthalpy of 391.2 kJ/kg . The gas leaves the compressor at a temperature of 245°C , a pressure of 0.6 MPa , and an enthalpy of 534.5 kJ/kg . There is no heat transfer to or from the gas as it flows through the compressor. (a) Evaluate the external work done per unit mass of gas assuming the gas velocities at entry and exit to be negligible. (b) Evaluate the external work done per unit mass of gas when the gas velocity at entry is 80 m/s and that at exit is 160 m/s .

Ans. 143.3 kJ/kg , 152.9 kJ/kg

- 5.11 The steam supply to an engine comprises two streams which mix before entering the engine. One stream is supplied at the rate of 0.01 kg/s with an enthalpy of 2952 kJ/kg and a velocity of 20 m/s . The other stream is supplied at the rate of 0.1 kg/s with an enthalpy of 2569 kJ/kg and a velocity of 120 m/s . At the exit

from the engine the fluid leaves as two streams, one of water at the rate of 0.001 kg/s with an enthalpy of 420 kJ/kg and the other of steam; the fluid velocities at the exit are negligible. The engine develops a shaft power of 25 kW. The heat transfer is negligible. Evaluate the enthalpy of the second exit stream.

Ans. 2401 kJ/kg

- 5.12 The stream of air and gasoline vapour, in the ratio of 14:1 by mass, enters a gasoline engine at a temperature of 30°C and leaves as combustion products at a temperature of 790°C. The engine has a specific fuel consumption of 0.3 kg/kWh. The net heat transfer rate from the fuel-air stream to the jacket cooling water and to the surroundings is 35 kW. The shaft power delivered by the engine is 26 kW. Compute the increase in the specific enthalpy of the fuel-air stream, assuming the changes in kinetic energy and in elevation to be negligible.

Ans. - 1877 kJ/kg mixture

- 5.13 An air turbine forms part of an aircraft refrigerating plant. Air at a pressure of 295 kPa and a temperature of 58°C flows steadily into the turbine with a velocity of 45 m/s. The air leaves the turbine at a pressure of 115 kPa, a temperature of 2°C, and a velocity of 150 m/s. The shaft work delivered by the turbine is 54 kJ/kg of air. Neglecting changes in elevation, determine the magnitude and sign of the heat transfer per unit mass of air flowing. For air, take $c_p = 1.005$ kJ/kg K and the enthalpy $h = c_p t$.

Ans. + 7.96 kJ/kg

- 5.14 In a turbomachine handling an incompressible fluid with a density of 1000 kg/m³ the conditions of the fluid at the rotor entry and exit are as given below

	Inlet	Exit
Pressure	1.15 MPa	0.05 MPa
Velocity	30 m/s	15.5 m/s
Height above datum	10 m	2 m

If the volume flow rate of the fluid is 40 m³/s, estimate the net energy transfer from the fluid as work.

Ans. 60.3 MW

- 5.15 A room for four persons has two fans, each consuming 0.18 kW power, and three 100 W lamps. Ventilation air at the rate of 80 kg/h enters with an enthalpy of 84 kJ/kg and leaves with an enthalpy of 59 kJ/kg. If each person puts out heat at the rate of 630 kJ/h determine the rate at which heat is to be removed by a room cooler, so that a steady state is maintained in the room.

Ans. 1.92 kW

- 5.16 Air flows steadily at the rate of 0.4 kg/s through an air compressor, entering at 6 m/s with a pressure of 1 bar and a specific volume of 0.85 m³/kg, and leaving at 4.5 m/s with a pressure of 6.9 bar and a specific volume of 0.16 m³/kg. The internal energy of the air leaving is 88 kJ/kg greater than that of the air entering. Cooling water in a jacket, surrounding the cylinder absorbs heat from the air at the rate of 59 W. Calculate the power required to drive the compressor and the inlet and outlet cross-sectional areas.

(Ans. 45.4 kW, 0.057 m², 0.0142 m²)

- 5.17 Steam flowing in a pipeline is at a steady state represented by p_p, t_p, u_p, v_p, h_p and V_p . A small amount of the total flow is led through a small tube to an evacuated chamber which is allowed to fill slowly until the pressure is equal to the pipeline

pressure. If there is no heat transfer, derive an expression for the final specific internal energy in the chamber, in terms of the properties in the pipeline.

- 5.18 The internal energy of air is given, at ordinary temperatures, by

$$u = u_0 + 0.718 t$$

where u is in kJ/kg, u_0 is any arbitrary value of u at 0°C , kJ/kg, and t is temperature in $^\circ\text{C}$.

Also for air, $p v = 0.287 (t + 273)$

where p is in kPa and v is in m^3/kg .

- (a) An evacuated bottle is fitted with a valve through which air from the atmosphere, at 760 mm Hg and 25°C , is allowed to flow slowly to fill the bottle. If no heat is transferred to or from the air in the bottle, what will its temperature be when the pressure in the bottle reaches 760 mm Hg?

Ans. 144.2°C

- (b) If the bottle initially contains 0.03 m^3 of air at 400 mm Hg and 25°C , what will the temperature be when the pressure in the bottle reaches 760 mm Hg?

Ans. 71.6°C

- 5.19 A pressure cylinder of volume V contains air at pressure p_0 and temperature T_0 . It is to be filled from a compressed air line maintained at constant pressure p_1 and temperature T_1 . Show that the temperature of the air in the cylinder after it has been charged to the pressure of the line is given by

$$T = \frac{\gamma T_1}{1 + \frac{p_0}{p_1} \left(\gamma \frac{T_1}{T_0} - 1 \right)}$$

- 5.20 A small reciprocating vacuum pump having the rate of volume displacement V_d is used to evacuate a large vessel of volume V . The air in the vessel is maintained at a constant temperature T by energy transfer as heat. If the initial and final pressures are p_1 and p_2 respectively, find the time taken for the pressure drop and the necessary energy transfer as heat during evacuation. Assume that for air, $pV = mRT$, where m is the mass and R is a constant, and v is a function of T only.

$$\left[\text{Ans. } t = \frac{V}{V_d} \ln \frac{p_1}{p_2}; Q = (p_1 - p_2) V \right]$$

[Hint: $dm = -p(V_d dt)/(RT) = V dp/(RT)$].

- 5.21 A tank containing 45 kg of water initially at 45°C has one inlet and one exit with equal mass flow rates. Liquid water enters at 45°C and a mass flow rate of 270 kg/h. A cooling coil immersed in the water removes energy at a rate of 7.6 kW. The water is well mixed by a paddle wheel with a power input of 0.6 kW. The pressures at inlet and exit are equal. Ignoring changes in KE and PE, find the variation of water temperature with time.

Ans. $T = 318 - 22 [1 - \exp(-6t)]$

- 5.22 A rigid tank of volume 0.5 m^3 is initially evacuated. A tiny hole develops in the wall, and air from the surroundings at 1 bar, 21°C leaks in. Eventually, the pressure in the tank reaches 1 bar. The process occurs slowly enough that heat transfer between the tank and the surroundings keeps the temperature of the air inside the tank constant at 21°C . Determine the amount of heat transfer.

Ans. -50 kJ

Second Law of Thermodynamics

6.1 Qualitative Difference between Heat and Work

The first law of thermodynamics states that a certain energy balance will hold when a system undergoes a change of state or a thermodynamic process. But it does not give any information on whether that change of state or the process is at all feasible or not. The first law cannot indicate whether a metallic bar of uniform temperature can spontaneously become warmer at one end and cooler at the other. All that the law can state is that if this process did occur, the energy gained by one end would be exactly equal to that lost by the other. *It is the second law of thermodynamics which provides the criterion as to the probability of various processes.*

Spontaneous processes in nature occur only in one direction. Heat always flows from a body at a higher temperature to a body at a lower temperature, water always flows downward, time always flows in the forward direction. The reverse of these never happens spontaneously. The spontaneity of the process is due to a finite driving potential, sometimes called the 'force' or the 'cause', and what happens is called the 'flux', the 'current' or the 'effect'. The typical forces like temperature gradient, concentration gradient, and electric potential gradient, have their respective conjugate fluxes of heat transfer, mass transfer, and flow of electric current. These transfer processes can never spontaneously occur from a lower to a higher potential. This directional law puts a limitation on energy transformation other than that imposed by the first law.

Joule's experiments (Article 4.1) amply demonstrate that energy, when supplied to a system in the form of work, can be completely converted into heat (work transfer $\rightarrow$ internal energy increase $\rightarrow$ heat transfer). But the complete conversion of heat into work in a cycle is not possible. So *heat and work are not completely interchangeable forms of energy.*

When work is converted into heat, we always have

$$W \Rightarrow Q$$

but when heat is converted into work in a complete closed cycle process

$$Q \not\Rightarrow W$$

The arrow indicates the direction of energy transformation. This is illustrated in Fig. 6.1. As shown in Fig. 6.1(a), a system is taken from state 1 to state 2 by work transfer W_{1-2} , and then by heat transfer Q_{2-1} the system is brought back from state 2 to state 1 to complete a cycle. It is always found that $W_{1-2} = Q_{2-1}$. But if the system is taken from state 1 to state 2 by heat transfer Q_{1-2} , as shown in Fig. 6.1(b), then the system cannot be brought back from state 2 to state 1 by work transfer W_{2-1} . Hence, heat cannot be converted completely and continuously into work in a cycle. Some heat has to be rejected. In Fig. 6.1(b), W_{2-1} is the work done and Q_{1-2} is the heat rejected to complete the cycle. This underlies the work of Sadi Carnot, a French military engineer, who first studied this aspect of energy transformation (1824). Work is said to be a *high grade energy* and heat a *low grade energy*. The complete conversion of low grade energy into high grade energy in a cycle is impossible.

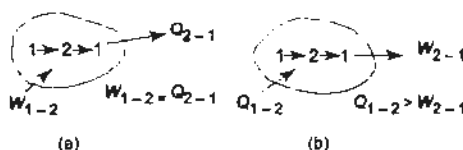


Fig. 6.1 Qualitative distinction between heat and work

6.2 Cyclic Heat Engine

For engineering purposes, the second law is best expressed in terms of the conditions which govern the production of work by a thermodynamic system operating in a cycle.

A heat engine cycle is a thermodynamic cycle in which there is a net heat transfer to the system and a net work transfer from the system. The system which executes a heat engine cycle is called a *heat engine*.

A heat engine may be in the form of a mass of gas confined in a cylinder and piston machine (Fig. 6.2a) or a mass of water moving in a steady flow through a steam power plant (Fig. 6.2b).

In the cyclic heat engine, as represented in Fig. 6.2(a), heat Q_1 is transferred to the system, work W_e is done by the system, work W_c is done upon the system, and then heat Q_2 is rejected from the system. The system is brought back to the initial state through all these four successive processes which constitute a heat engine cycle. In Fig. 6.2(b) heat Q_1 is transferred from the furnace to the water in the